

Network Tariff Guide for Standard Control Services



1 July 2014 to 30 June 2015



Revision history

Version	Date	Summary of changes
1.0	17 June 2014	Initial 2014–15 Network Tariff Guide for Standard Control Services
1.1	27 June 2014	Minor amendments to Table 6.1, Table 8.7 and Section 8.4.4.
1.2	14 August 2014	Reference to 'Minimum chargeable demand' changed to 'Threshold above which demand charge applies' in Appendix 2

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1 Introduction

1.1 Overview

Ergon Energy Corporation Limited (Ergon Energy) is a Distribution Network Service Provider (DNSP) to over 700,000 customers in regional Queensland. Our service area covers around 97 per cent of Queensland and has approximately 150,000 kilometres of power lines and one million power poles. Around 70 per cent of the network's power lines are radial and service mostly rural areas with very low levels of customers per line kilometre.

As a DNSP, Ergon Energy is subject to economic regulation by the Australian Energy Regulator (AER) under the National Electricity Law (the Law) and the National Electricity Rules (NER). Under the Law and NER, the AER is responsible for regulating the revenues Ergon Energy can earn, and the prices that we can charge for certain services provided by means of, or in connection with, our distribution system.

In order to regulate the prices that Ergon Energy can charge for our services, the AER has classified our regulated distribution services into Standard Control Services and Alternative Control Services.

Standard Control Services are core network, connection and metering services associated with the access and supply of electricity to customers (e.g. conveying electricity, network maintenance, installation of standard meters and load control equipment etc.). Ergon Energy recovers our costs in providing Standard Control Services through network tariffs which are billed to retailers.

Alternative Control Services are comprised of:

- *Fee Based Services* – regulated distribution activities which Ergon Energy undertakes at the request of an identifiable customer, retailer or appropriate third party which are in addition to our main network, connection and metering services and are levied as a separate charge. Fee Based Services have a 'fixed fee' as the costs of providing the service (and therefore price) can be assessed in advance of the service being requested by a customer or retailer (e.g. de-energisations, re-energisations, and supply abolishment etc.).
- *Quoted Services* – similar to Fee Based Services, but they are 'priced on application' as the nature and scope of these services are variable and the costs (and therefore price) are specific to the individual retailer's or customer's needs (e.g. Large Customer Connections, and emergency recoverable works etc.).
- *ACS – Street Lighting Services* – relate to the installation, operation and maintenance of street lighting assets owned by Ergon Energy. Ergon Energy recovers our costs in providing ACS – Street Lighting Services through a daily street lighting charge which we bill to retailers.

1.2 Purpose and structure

This Tariff Guide sets out the Network Tariff Codes and standard rates that apply in 2014–15 for Standard Control Services, as well as the application rules. It is structured as follows:

- Section 2 sets out general requirements that apply to the selection and amendment of network tariffs
- Section 3 outlines the tariff classes and Network Tariff Codes, as well as the procedures for assigning and reassigning customers to tariff classes and network tariffs
- Section 4 describes the four network user groups used by Ergon Energy when setting network tariffs

- Section 5 describes and depicts the three pricing zones used by Ergon Energy when setting network tariffs
- Section 6 details the standard 2014–15 Distribution Loss Factors (DLFs)
- Section 7 provides information on the Transmission Node Identifier (TNI)
- Section 8 describes the Distribution Use of System (DUOS) and Transmission Use of System (TUOS) tariffs applying to each network user group. This section also includes information on the Queensland Government Solar Bonus Scheme.

Appendix 2 sets out the standard rates that apply in 2014–15. Information on Network Tariff Codes and rates applying to site-specific tariffs are available on request by contacting Network Pricing (see Section 1.5). Ergon Energy will only release this information in the following circumstances:

- the requestor is the customer, or an authorised person who is able to sign on behalf of the customer (e.g. Company Manager)
- the requestor is the current Financially Responsible Market Participant and the request relates to a period of time for which they are responsible for the site
- the customer's written consent has been provided using the "Authority to Release Information to Third Parties Form" available on Ergon Energy's website (see Section 1.4).

Finally, Appendix 3 provides the methodology used by Ergon Energy to calculate our Inclining Block Tariff (IBT) network tariffs for Standard Asset Customers (SAC) <100 MWh per annum (p.a.) and Appendix 4 details the kVA methodology used to calculate tariffs for Individually Calculated Customers (ICCs).

1.3 Changes in 2014–15

Ergon Energy is restructuring the way we charge for the use of our distribution network to help ensure we maintain a viable network for our customers into the future. This process is expected to take a number of years, with the first changes occurring in 2014–15.

In broad terms, Ergon Energy has made the following changes to network tariffs in 2014–15:

- introduced kVA as the basis for the demand tariffs used by our very large energy users
- commenced the process of 'rebalancing' tariffs towards more fixed/less usage-dependent charges for our large and small users
- introduced new and optional broad-based tariff structures for our smaller customers.

The tariff structures are designed to allow our customers, through their retail account, to better understand the cost associated with accessing and using the network.

Further information on these changes is contained in our 2014–15 Pricing Proposal¹ and on our website at: www.ergon.com.au/futurenetworktariffs.

1.4 Supporting network pricing documents

In addition to this Tariff Guide, Ergon Energy has a number of network pricing documents to assist customers, retailers and interested parties understand how network tariffs and connection charges are developed and applied. These documents are outlined in Figure 1.1 below and are available on our website at: www.ergon.com.au/networktariffs.

¹ See www.ergon.com.au/networktariffs.

Figure 1.1: Supporting network pricing documentation



1.5 Further information

Network users and retailers who are uncertain about the network pricing process or their particular circumstances are encouraged to contact us for assistance:

Manager Regulatory Determination and Pricing
 PO Box 264
 FORTITUDE VALLEY QLD 4006
 Phone: 13 10 46
 E-mail: netprice@ergon.com.au.

2 General requirements

The following general requirements apply to the selection and amendment of network tariffs:

- It is intended that a Network Tariff Code will apply to each meter data stream.
- Network users with multiple network connections will pay network charges for each connection point. This approach is consistent with the National Metering Identifier (NMI) Procedure issued by the Australian Energy Market Operator (AEMO).
- If requested, Ergon Energy will provide the customer or the customer's retailer with reasonable assistance in identifying the network tariffs which may be applicable to the customer's installation.
- Demand and/or volume are to be measured at each individual network connection point.
- Where applicable, demand based charges will be calculated on a pro rata daily basis for initial and final accounts.
- Where Ergon Energy's load control equipment exists (including associated metering), this may not be disconnected without Ergon Energy's prior written consent.
- Residential and Business customer classifications align with the Queensland Electricity Industry Code.
- In the absence of any other customer classification indicators, the IBT Business tariff will be the default primary tariff for SACs <100 MWh p.a.
- The Time-of-Use (TOU) tariffs² are only available to SACs on a market retail contract who consume less than 100 MWh p.a.
- Where a retailer wants to access the network TOU tariff, and is offering a retail TOU tariff that differs in the Peak, Shoulder and Off-Peak times from the applicable network TOU tariff, then this will be considered a special arrangement. The costs of this special arrangement will be the retailer's responsibility.
- For the purposes of network billing for the IBT, the energy (kWh) assigned to each consumption block during a meter reading period are pro-rated back to a daily equivalent with the tariff applying on a daily basis. The IBT network tariff calculation methodology is set out in Appendix 3.
- Network charges will be billed by Ergon Energy on a monthly basis.

² TOU Residential and TOU Business

3 Tariff classes

Ergon Energy's tariffs are currently differentiated at the highest level by:

- network user groups:
 - ICCs
 - Connection Asset Customers (CACs)
 - SACs
 - Embedded Generators (EGs)
- zones:
 - East Zone
 - West Zone
 - Mount Isa Zone
- for large customers, whether the customer has connected under the previous regulatory arrangements where connection assets form part of the network tariff or the Large Customer Connection arrangements where new or augmented connection assets are paid for or contributed by the customer.

A list of Ergon Energy's tariff classes and corresponding Network Tariff Codes for our Standard Control Services are set out in Table 3.1 below.

Table 3.1: Tariff classes and Network Tariff Codes

Tariff class	Tariff	Network Tariff Codes
Individually Calculated Customer (Pre 30 June 2010) – East	ICC	EICCKA1 onwards
Individually Calculated Customer (Pre 30 June 2010) – West	ICC	WICCKA1 onwards
Individually Calculated Customer (Pre 30 June 2010) – Mount Isa	ICC	MICCKA1 onwards
Individually Calculated Customer (Post 30 June 2010) – East	ICC	EICCKB1 onwards
Individually Calculated Customer (Post 30 June 2010) – West	ICC	WICCKB1 onwards
Individually Calculated Customer (Post 30 June 2010) – Mount Isa	ICC	MICCKB1 onwards
Connection Asset Customers (Pre 30 June 2010) – East	CAC	ECACA1 onwards
Connection Asset Customers (Pre 30 June 2010) – West	CAC	WCACA1 onwards
Connection Asset Customers (Pre 30 June 2010) – Mount Isa	CAC	MCACA1 onwards
Connection Asset Customers (Post 30 June 2010) – East	CAC	ECACB1 onwards
Connection Asset Customers (Post 30 June 2010) – West	CAC	WCACB1 onwards
Connection Asset Customers (Post 30 June 2010) – Mount Isa	CAC	MCACB1 onwards

Tariff class	Tariff	Network Tariff Codes
Embedded Generation (Pre 30 June 2010) – East	EG	EEGA1 onwards
Embedded Generation (Pre 30 June 2010) – West	EG	WEGA01 onwards
Embedded Generation (Pre 30 June 2010) – Mount Isa	EG	MEGA01 onwards
Embedded Generation (Post 30 June 2010) – East	EG	EEGB1 onwards
Embedded Generation (Post 30 June 2010) – West	EG	WEGB01 onwards
Embedded Generation (Post 30 June 2010) – Mount Isa	EG	MEGB01 onwards
Standard Asset Customer – Large (>100 MWh p.a.) – East	Demand High Voltage	EDHTT1, EDHTT2, EDHTT3
	Demand Large	EDLTT1, EDLTT2, EDLTT3
	Demand Medium	EDMTT1, EDMTT2, EDMTT3
	Demand Small	EDSTT1, EDSTT2, EDSTT3
Standard Asset Customer – Large (>100 MWh p.a.) – West	Demand High Voltage	WDHTT1, WDHTT2, WDHTT3
	Demand Large	WDLTT1, WDLTT2, WDLTT3
	Demand Medium	WDMTT1, WDMTT2, WDMTT3
	Demand Small	WDSTT1, WDSTT2, WDSTT3
Standard Asset Customer – Large (>100 MWh p.a.) – Mount Isa	Demand High Voltage	MIDHT
	Demand Large	MIDLT
	Demand Medium	MIDMT
	Demand Small	MIDST
Standard Asset Customer – Small (<100 MWh p.a.) – East	IBT Business	EBIBT1, EBIBT2, EBIBT3
	TOU Business	EBTOUT1, EBTOUT2, EBTOUT3
	IBT Residential	ERIBT1, ERIBT2, ERIBT3
	TOU Residential	ERTOUT1, ERTOUT2, ERTOUT3
	Volume Controlled	EVCT1, EVCT2, EVCT3
	Volume Night Controlled	EVNT1, EVNT2, EVNT3
Standard Asset Customer – Small (<100 MWh p.a.) – West	IBT Business	WBIBT1, WBIBT2, WBIBT3
	TOU Business	WBTOUT1, WBTOUT2, WBTOUT3
	IBT Residential	WRIBT1, WRIBT2, WRIBT3
	TOU Residential	WRTOUT1, WRTOUT2, WRTOUT3
	Volume Controlled	WVCT1, WVCT2, WVCT3
	Volume Night Controlled	WVNT1, WVNT2, WVNT3
Standard Asset Customer – Small (<100 MWh p.a.) – Mount Isa	IBT Business	MIBIB
	TOU Business	MIBTOU
	IBT Residential	MIRIB
	TOU Residential	MIRTOU
	Volume Controlled	MIVC
	Volume Night Controlled	MIVN
Standard Asset Customer – Unmetered – East	Volume Unmetered	EVUT1, EVUT2, EVUT3 EVUT1MI, EVUT2MI, EVUT3MI EVUT1MA, EVUT2MA, EVUT3MA

Tariff class	Tariff	Network Tariff Codes
Standard Asset Customer – Unmetered – West	Volume Unmetered	WVUT1, WVUT2, WVUT3 WVUT1MI, WVUT2MI, WVUT3MI WVUT1MA, WVUT2MA, WVUT3MA
Standard Asset Customer – Unmetered – Mount Isa	Volume Unmetered	MIVU, MIVUMI, MIVUMA

3.1 Assignment and reassignment of customers to tariff classes

The AER requires Ergon Energy to comply with the procedures for assigning or reassigning customers to tariff classes³ as per the requirements set out in its Final Distribution Determination.

Assignment or reassignment of customers to Ergon Energy's tariff classes can occur as a result of:

- customers requesting a new connection to the network
- existing customers applying for increased capacity on the network
- a change in a customer's NMI classification
- annual review as part of Ergon Energy's process of developing and submitting a Pricing Proposal for approval by the AER
- requests for a review of the assigned tariff class by either a customer and/or retailer.

3.2 Notification of a tariff class assignment and reassignment

From 1 July 2010, the AER requires Ergon Energy to notify the customer concerned of the tariff class we will assign the customer to, prior to a new assignment or reassignment occurring.⁴ This information will be made available to the customer and retailer via a written notification.

The effective date for a Standard Control Services tariff class assignment or reassignment will be the date as advised by Ergon Energy.

It is important to note that a tariff class reassignment may necessitate a change to a customer's network tariff assigned to a NMI. Details of Ergon Energy's proposed new network tariffs will be included in the written notification mentioned above. Subject to any appeal, the new tariff class and new Network Tariff Code will take effect from the date advised by Ergon Energy in the written notification.

A customer or retailer may request further information relating to a particular tariff class assignment or reassignment decision by contacting Network Pricing (see Section 1.5).

3.3 Review of a customer's assigned tariff class

Ergon Energy does not reassign customers to tariff classes without careful review and adequate justification. Reassignment of a customer's tariff class generally only occurs in a situation where a customer alters the underlying characteristics of their connection, in terms of size or nature of usage.

In addition to Ergon Energy's own review of tariff classes applying to customers, a customer and/or retailer may request a review of a customer's assigned tariff class at any time during a regulatory year. This

³ A tariff class is defined under the NER and relates to a group or class of customers who are subject to a particular tariff or tariffs.

⁴ Customers connected to our network prior to 1 July 2010 are deemed to be 'assigned' to the tariff class which we were charging the customer immediately prior to 1 July 2010.

request may occur directly through an explicit request to review a tariff class assigned to a customer, or indirectly through a request to change a network tariff assigned to a NMI.

Ergon Energy relies on a range of information (e.g. historical consumption data) and has specific criteria for assessing the reassignment of customers to tariff classes. The criteria, by network user group, are set out in Table 3.2 below. Once a customer is identified for reassignment, the connection characteristics and the customer's expected energy consumption are used to determine the appropriate customer group, and hence tariff class, to which the customer should be reassigned.

Table 3.2: Tariff class reassignment criteria for Standard Control Services

Network user group	Typical characteristics of customers assigned	Criteria for reassigning customers to a different tariff class
SACs	Annual consumption is expected to be below 4 GWh p.a.	- Annual consumption increases, or is expected to increase, above 4 GWh p.a., and/or - A customer requests an increase in supply capacity requiring augmentation to connection assets which results in a dedicated supply system which is quite different and separate from the remainder of the supply network.
CACs	- Required capacity above 1,500 kVA, or - Annual consumption is expected to exceed 4 GWh p.a.	Reassigned to ICCs: - Annual consumption increases, or is expected to increase, above 40 GWh p.a., and/or - A customer requests an increase in supply capacity requiring augmentation to their connection assets which results in a dedicated supply system which is quite different and separate from the remainder of the supply network. Reassigned to SACs: - Annual consumption reduces or is expected to reduce below 4 GWh p.a. and their dedicated supply system is not considered to be quite different and separate from the remainder of the supply network, and/or - Required capacity falls below 1,500 kVA.⁵
ICCs	- Annual consumption is expected to exceed 40 GWh p.a., or - Their dedicated supply system is considered to be quite different and separate from the remainder of the supply network.	Annual consumption reduces or is expected to reduce below 40 GWh p.a. and their dedicated supply system is considered comparable with CACs at the same voltage level.

The following general requirements apply to the review and amendment of a customer's tariff class assignment for Standard Control Services:

- Any change in an assigned tariff class will only proceed upon approval by Ergon Energy. Ergon Energy will make our determination having regard to this Tariff Guide, our Tariff Class Reassignment Criteria (as set out above), the AER's Final Distribution Determination and any applicable law.

⁵ With the exception of those customers who have a dedicated supply system which is quite different and separate from the remainder of the supply network or where inequitable treatment of otherwise comparable customers will arise from the application of the 4 GWh threshold.

- Requests should be made by the customer, and are preferably made to Ergon Energy through the customer's retailer. Requests for review of a tariff class within the SAC network user group (e.g. from SAC >100 MWh p.a. to SAC <100 MWh p.a.) should be emailed to the Service Transaction Centre (servicetransactioncentre@ergon.com.au). All other requests for a tariff class review should be emailed to Network Pricing (refer to Section 1.5).
- Subject to a change in the customer's usage or pattern of usage at the customer's premises, a further request to review and change a tariff class should not be made until a period of 12 months has elapsed from the previously approved request.
- Any change in the tariff class, and associated network tariff(s) will take effect from the date as approved by Ergon Energy (the effective date). Network charges under an existing network tariff within an existing tariff class will continue to be billed by Ergon Energy until the effective date. Backdating of charges prior to the effective date will not be allowed by Ergon Energy.

3.4 Objections to a tariff class assignment or reassignment

If a customer or a retailer raises an objection to a tariff class assignment or reassignment, Ergon Energy will make every effort to investigate and address the customer's or retailer's concerns when the customer or retailer firsts contacts Ergon Energy.

In the first instance, the customer or retailer should contact Network Pricing via e-mail requesting a review of Ergon Energy's proposed tariff class assignment or reassignment. The request should outline:

- the customer's name, NMI and contact details (or retailer's contact details if acting on behalf of the customer)
- what the customer or the retailer is objecting to (noting Ergon Energy's decision is made having regard to this Tariff Guide, our Tariff Class Reassignment Criteria, the AER's Final Distribution Determination and any applicable law), and
- what outcome the customer or retailer is seeking.

Ergon Energy will endeavour to inform the customer or the retailer of the outcome of our review, including a reason for our decision within five business days of receiving such a request.

If the customer or the retailer is not satisfied with the outcome of this review, the objection can be escalated to the Manager Regulatory Determination and Pricing. The customer or the retailer can request an objection to be escalated by contacting Network Pricing via e-mail. The Manager Regulatory Determination and Pricing will advise the customer or the retailer of the outcome of an escalated objection as soon as practical.

If Ergon Energy is unable to resolve the objection to the customer's satisfaction, the customer can refer the matter to the AER for resolution via the dispute resolution process available under Part 10 of the Law.

3.5 Network tariff reviews

The customer or the retailer may apply to Ergon Energy for a change to the network tariff assigned to a NMI by sending a request to the Service Transaction Centre via e-mail (servicetransactioncentre@ergon.com.au).

Ergon Energy will inform the customer and/or retailer within five business days of receiving such a request, of the decision taken.

If Ergon Energy:

- agrees that the network tariff proposed by the customer or retailer is applicable to the NMI, Ergon Energy will advise the retailer of the effective date of application, or
- disagrees that the network tariff proposed by the customer or retailer is applicable to the NMI, Ergon Energy will notify the customer and/or retailer, including a reason for the rejection.

A customer and/or retailer may appeal the network tariff decision. The appeals process to follow is outlined in Table 3.3 below.

Table 3.3: Network tariff appeals process

Scenario	Appeals process
The customer and/or retailer does not agree with Ergon Energy's decision to maintain the current network tariff.	<u>If it relates to a change to a network tariff within the SAC network user group:</u> ▪ A request to review the decision should be sent to the Service Transaction Centre via email. This request should be sent by the customer's retailer. <u>Otherwise:</u> ▪ A request to review the decision should be sent to the Network Pricing via email. This request should be sent by the customer's retailer.
The customer does not agree to a network tariff change initiated by their retailer.	▪ The customer should contact their retailer in the first instance. ▪ Following this, if the matter is not resolved to the satisfaction of the customer, the customer may submit a request to Ergon Energy to review the network tariff decision. ○ If it relates to a change to a network tariff within the SAC network user group, this request should be sent to the Service Transaction Centre via email. ○ Otherwise, this request should be sent to Network Pricing via email.

Ergon Energy will review an appeal and the customer's eligibility for a network tariff with regard to this Tariff Guide, our Tariff Class Reassignment Criteria, the AER's Final Distribution Determination and any applicable law. Ergon Energy will advise the customer and/or the retailer of the outcome of an appeals review as soon as practical.

Any change in a network tariff assigned to a NMI will take effect from the date as approved by Ergon Energy (the effective date). Network charges under the existing network tariff will continue to be billed by Ergon Energy until the effective date. Backdating of charges prior to the effective date will not be allowed by Ergon Energy.

Subject to a change in usage or the pattern of usage at the customer's installation, a further request to change the Network Tariff Code assigned to a NMI should not be made until a period of 12 months has elapsed from the previously approved request for the NMI.

If Ergon Energy agrees with a proposed network tariff and determines that a tariff class reassignment is also required, written notification of the new tariff class and new Network Tariff Code will also be provided to the customer and retailer prior to the tariff class reassignment occurring, in accordance with Section 3.3.

4 Network user groups

As noted above, Ergon Energy has four network user groups (with multiple tariff classes within these groups). A description of the four network user groups is provided in Table 4.1 below.

Table 4.1: Ergon Energy's network user groups

Network user group	Description
ICC	Those customers: - with energy consumption typically greater than 40 GWh p.a., or - with energy consumption lower than 40 GWh p.a. where: - a customer has a dedicated supply system which is quite different and separate from the remainder of the supply network - there are only two or three customers in a supply system making average prices inappropriate - a customer is connected at or close to a Transmission Network Connection Point (TNCP), or - inequitable treatment of otherwise comparable customers will arise from the application of the 40 GWh p.a. threshold.
CAC	Those customers: - with required capacity above 1,500 kVA - with energy consumption typically greater than 4 GWh p.a., or - with required capacity below 1,500 kVA where: - a customer has a dedicated supply system which is quite different and separate from the remainder of the supply network, or - inequitable treatment of otherwise comparable customers will arise from the application of the 4 GWh p.a. threshold. The CAC group is further subdivided into categories based on voltage levels as follows: 66 kV – connected to either a 66 kV substation or a 66 kV line 33 kV – connected to either a 33 kV substation or a 33 kV line 22/11 kV Bus – connected to either a 22 kV or 11 kV substation 22/11 kV Line – connected to either a 22 kV or 11 kV line.
SAC	All other load customers. This includes customers with micro generation facilities (such as small scale photovoltaic (PV) generators) that have a similar service connection and usage profile as other SACs without such facilities. Customers taking supply at LV with a generator installed with a capacity less than or equal to 1 MW, will generally be classified as a SAC unless the connection assets are considered to be quite different from that supplying other SACs. The SAC group is further subdivided into network tariff categories based on whether: - the customer's connection is metered or unmetered - the customer's consumption relates to residential or business use - the customer is taking supply at high voltage or low voltage - the customer's consumption is above or below 100 MWh p.a. - the customer has a meter installed capable of recording demand - the customer's supply is capable of being controlled by Ergon Energy.
EG	Those network users that export energy into the distribution system. EGs do not include customers with micro generation facilities that have been classified as a SAC. EGs are separated into two categories: - EGs that are connected to the distribution system and only generate into the distribution system, or - EGs that are connected to the distribution system, generate and take load from the system.

5 Zones

5.1 Number of zones

Three pricing zones have been delineated in the Ergon Energy area of supply, broadly based on Queensland's local government areas (LGAs), with the distribution network electrical connection being the final determinant of which zone applies. Zone pricing impacts DUOS prices only; TUOS prices are not impacted by zones.

The three pricing zones⁶ are:

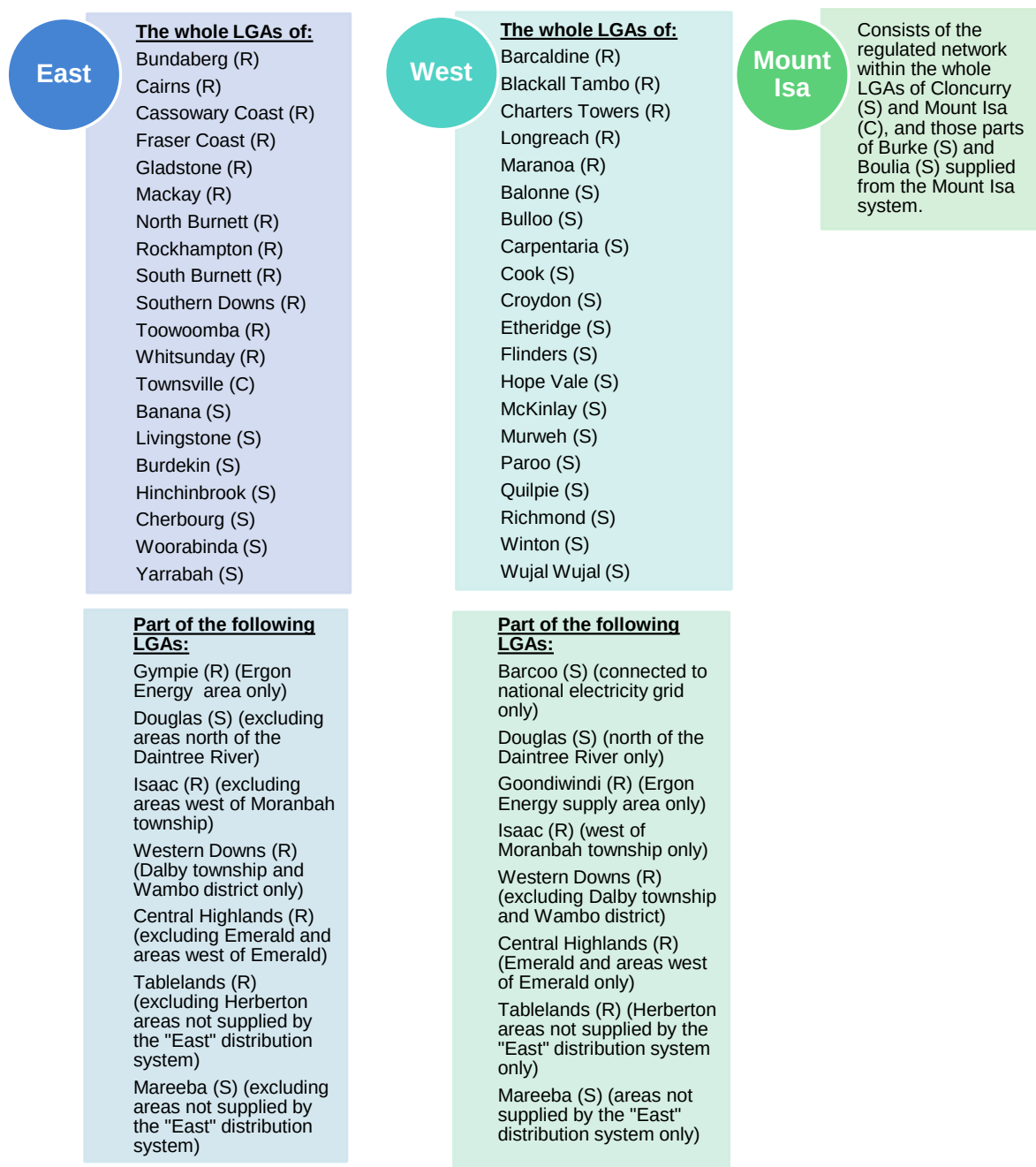
- **East Zone** – those areas where the network users are supplied from the distribution system connected to the national grid and have a relatively low distribution cost to supply
- **West Zone** – those areas outside the East Zone and connected to the national grid, which have a significantly higher distribution cost of supply than the East Zone
- **Mount Isa Zone** – broadly defined as those areas supplied from the isolated Mount Isa system. This zone is not connected to the national grid and as such would normally be excluded from the application of the NER. However, under the *Electricity – National Scheme (Queensland) Act 1997*, the Queensland Government has transferred responsibility for the economic regulation of the Mount Isa–Cloncurry supply network to the AER.

5.2 Zone coverage

The LGAs covered by each zone are shown in Figure 5.1 below.

⁶ Areas supplied from isolated (remote) generation are not included in any of the below zones.

Figure 5.1: Zone coverage

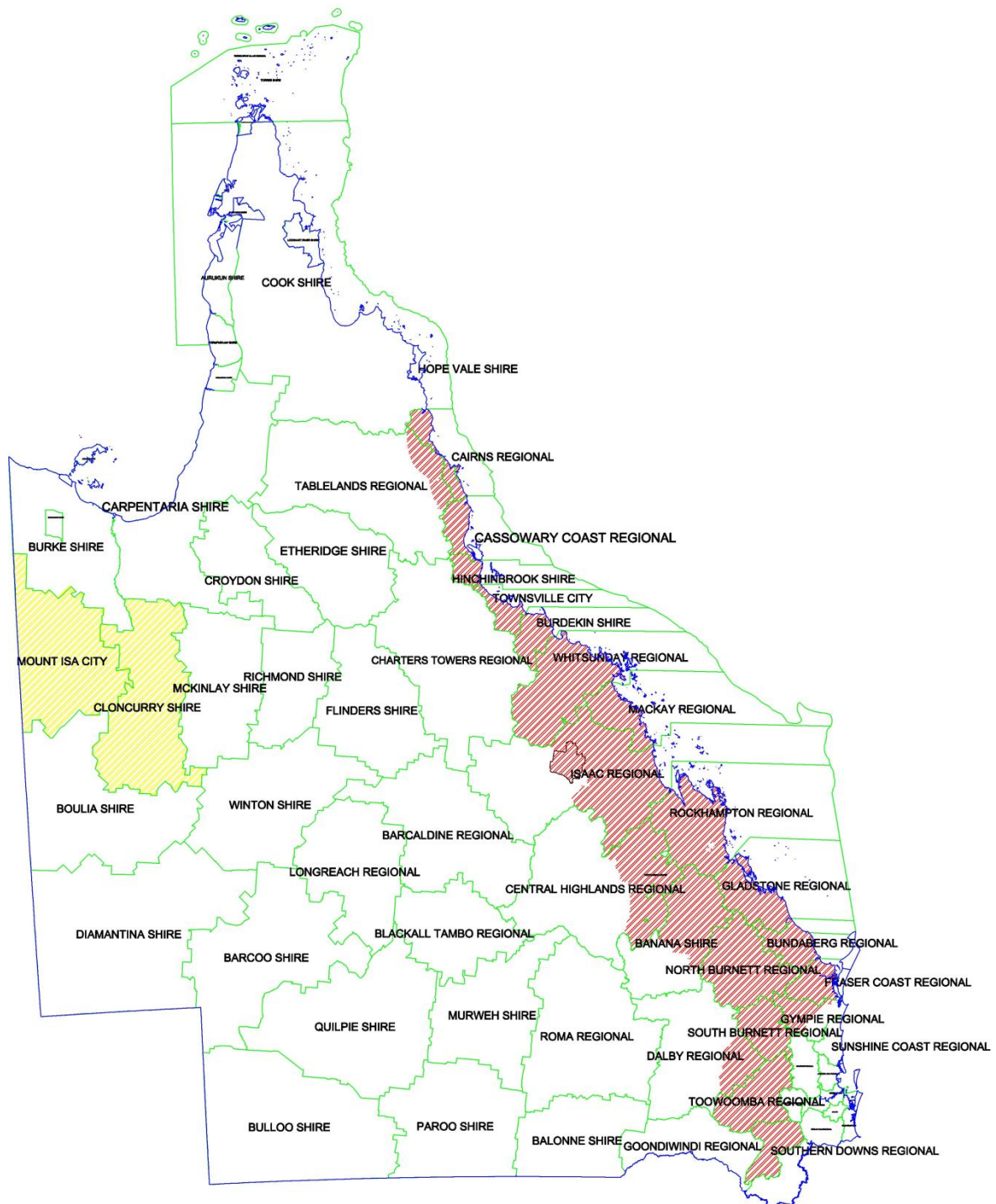


Note: (R) = Regional Council, (S) = Shire Council and (C) = City Council.

5.3 Zone map

The following map depicts the East Zone (red), the Mount Isa Zone (yellow) and the West Zone (the remaining area of the state supplied from the interconnected grid).

Figure 5.2: Zone map



6 Distribution Loss Factors

6.1 Overview

DLFs are calculated annually by Ergon Energy in accordance with requirements of the NER in order to determine the amount of energy dispatched to supply customers. They are approved by the AER and published by AEMO on their website.

Every NMI has a DLF code which is associated with the location of the metering point.

The DLF is a multiplier used to convert the actual metered energy into the equivalent energy passing through the appropriate TNCP by allowing for the distribution network losses that are incurred between the meter and the TNCP.

The DLF is applied to the metered consumption for the calculation of TUOS volume charges. DLFs are generally assigned on the basis of the standard metering voltage for the type of connection. However, a specific loss factor may be applied where there is a unique network supply configuration.

6.2 DLF categories and applications

Table 6.1 outlines the DLF categories used by Ergon Energy and their applications.

Table 6.1: DLF categories and applications

Category	Description	Application
Sub-transmission Bus	Applicable to connection points that are High Voltage metered at a Sub-transmission Bus with a voltage greater than 30 kV.	NMIs that are supplied from a zone substation by dedicated, greater than 30 kV lines and metered at or immediately adjacent to the zone substation would be eligible for a Sub-transmission Bus DLF. Zone substations are defined as substations where the voltage level is stepped down from a voltage greater than 30 kV.
Sub-transmission Line	Applicable to connection points that are High Voltage metered at a Sub-transmission Line with a voltage greater than 30 kV.	The Sub-transmission Line DLF will apply to all NMIs that are: - connected to high voltage sub-transmission lines (greater than 30 kV) - metered at the same voltage as the line is energised. If customers believe that the NMI would qualify for a high voltage Bus DLF rather than a high voltage Line DLF they should submit details justifying their claim to Ergon Energy. As an example, NMIs supplied from a zone substation by dedicated 22/11 kV lines and are metered at or immediately adjacent to the zone substation would be eligible for a 22/11 kV Bus DLF.
22/11kV Bus	Applicable to connection points that are High Voltage metered at a 22/11 kV Bus with a voltage less than 30 kV and greater than 1,000 volts.	As per "Sub-transmission Bus" application above.

Category	Description	Application
22/11kV Line	Applicable to connection points that are High Voltage metered at a 22/11 kV Line with a voltage less than 30 kV and greater than 1,000 volts.	The 22/11 kV Line DLF will apply to all NMIs that are: - connected to high voltage distribution lines - metered at the same voltage as the line is energised.
Low Voltage (LV) Bus	Applicable to connection points that are Low Voltage metered at a Low Voltage Bus with a voltage less than 1,000 volts.	NMIs which are supplied from a distribution substation on their site (owned or leased) and metered at the low voltage level are eligible for a LV Bus DLF.
Low Voltage (LV) Line	Applicable to connection points that are Low Voltage metered at a Low Voltage Line with a voltage less than 1,000 volts.	All NMIs not covered by any of the above categories. If customers believe that the NMI would qualify for a LV Bus DLF rather than a LV Line DLF they should submit details justifying their claim to Ergon Energy.

6.3 DLFs to apply in 2014–15

The standard DLFs that apply in 2014–15 are detailed in Table 6.2 below.

Table 6.2: Ergon Energy's 2014–15 Standard Distribution Loss Factors

Network level	East Zone *		West Zone		Mount Isa Zone	
	Code	DLF	Code	DLF	Code	DLF
Sub-transmission Bus	GESB	1.005	GWSB	1.055	GMSB	1.001
Sub-transmission Line	GESL	1.011	GWSL	1.086	GMSL	1.005
22/11 kV Bus	GEHB	1.015	GWHB	1.093	GMHB	1.007
22/11 kV Line	GEHL	1.034	GWHL	1.127	GMHL	1.035
LV Bus	GELB	1.079	GWLB	1.174	GMLB	1.056
LV Line	GELL	1.094	GWLL	1.184	GMLL	1.060

* The zonal boundaries are identical to those used for network tariffs.

Some larger ICCs, CACs and EGs have site-specific DLFs for 2014–15. This means the DLFs shown in Table 6.2 do not apply to these customers.

The site-specific DLFs, as well as the standard DLFs, are published on AEMO's website: <http://www.aemo.com.au/electricityops/lossfactors.html>.

DLFs for all NMIs will be included in the Schedule of Network Tariffs issued to the customer and/or their retailer. For SACs, the customer or retailer must nominate the DLF appropriate to the connection. The customer or retailer may be required to justify any nominated DLF.

Where the voltage at the metering point is not listed, or there is doubt about the application of these rules, please contact Ergon Energy.

6.4 Further information

Further information on Ergon Energy's methodology for calculating DLFs is available on our website: www.ergon.com.au/about-us/the-electricity-industry/electricity-distribution-pricing-methodologies/distribution-loss-factor-calculation-methodology.

Detailed information about the purpose and application of DLFs may also be obtained from the AEMO website mentioned in Section 6.3 above.

7 Transmission Node Identifiers

The TNI identifies the TNCP from which a customer is supplied. The NMI Discovery process (Stage 2) will provide retailers with the TNI applicable to a NMI.

8 Tariff structures

The DUOS and TUOS tariffs applying to each network user group are set out in the following sections, with the standard rates (GST exclusive and GST inclusive) available in Appendix 2. DUOS and TUOS tariffs for ICCs, CACs and EGs are site-specific and are available upon request (see Section 1.2 for conditions).

8.1 Individually Calculated Customers

8.1.1 Changes to ICC tariffs

The key change for ICC tariffs in 2014–15 is a move from a kW basis for the two demand-related tariff parameters (capacity and actual demand) to a kVA basis for the DUOS component. TUOS recovery has also been aligned with the change to kVA demand denomination to minimise network tariff and billing complexity. Ergon Energy's kVA calculation methodology is set out in Appendix 4.

8.1.2 DUOS tariffs

DUOS tariffs for ICCs are based on:

- the actual dedicated connection assets utilised by the customer
- the customer's specifically identified portion of any shared distribution network utilised for the electricity supply
- whether the customer is subject to the pre or post 30 June 2010 arrangements for Large Customer Connections
- whether the customer is located in the East, West or Mount Isa Zone of Ergon Energy's network.

DUOS tariffs incorporate a fixed charge (\$/day), a capacity charge (\$/kVA of AD/month), an actual demand charge (\$/kVA/month) and a volume charge (\$/kWh).

Where the actual demand exceeds the authorised demand (AD) in any one month, the actual demand will be substituted for the AD in the calculation of the capacity charge billed for that month.

8.1.3 TUOS tariffs

TUOS charges are calculated and levied for each TNCP and are passed through on a reflective basis proportional to the AD of the customer and their energy consumption.

TUOS tariffs incorporate a fixed charge (\$/day), a capacity charge (\$/kVA of AD/month), a common service and general charge (\$/day), and a volume charge (\$/kWh).

Where the actual demand exceeds the AD in any one month, the actual demand will be substituted for the AD in the calculation of the capacity charge billed for that month.

To determine the total TUOS volume charge, the metered consumption must be multiplied by the customer's DLF and then applied to the TUOS \$/kWh rate.

8.2 Connection Asset Customers

8.2.1 Changes to CAC tariffs

There have been no structural changes for CAC tariffs in 2014–15.

8.2.2 DUOS tariffs

DUOS tariffs for CACs are based on:

- the actual dedicated connection assets utilised by the customer
- average charges for use of the shared network
- whether the customer is subject to the pre or post 30 June 2010 arrangements for Large Customer Connections
- whether the customer is located in the East, West or Mount Isa Zone of Ergon Energy's network.

DUOS tariffs incorporate a fixed charge (\$/day), a capacity charge (\$/kW of AD/month), an actual demand charge (\$/kW/month) and a volume charge (\$/kWh).

Where the actual demand exceeds the AD in any one month, the actual demand will be substituted for the AD in the calculation of the capacity charge billed for that month.

8.2.3 TUOS tariffs

TUOS charges are calculated and levied for each TNCP and are passed through on a reflective basis proportional to the AD of the customer and their energy consumption.

TUOS tariffs incorporate a fixed charge (\$/day), a capacity charge (\$/kW of AD/month), a common service and general charge (\$/day), and a volume charge (\$/kWh).

Where the actual demand exceeds the AD in any one month, the actual demand will be substituted for the AD in the calculation of the capacity charge billed for that month.

To determine the total TUOS volume charge, the metered consumption must be multiplied by the customer's DLF and then applied to the TUOS \$/kWh rate.

8.3 Standard Asset Customers >100 MWh p.a.

8.3.1 Definition

SACs >100 MWh p.a. are those customers with energy consumption greater than 100 MWh p.a. who are not otherwise classified as an ICC, CAC or EG.

8.3.2 Changes to SAC >100 MWh p.a. tariffs

There has been no structural change to these tariffs in 2014–15. However, Ergon Energy has commenced rebalancing revenue recovery away from the actual demand charge and towards the fixed charge to increase cost reflectivity. Also, the minimum monthly demand charge mechanism has now been incorporated in the fixed charge and offset by the introduction of a threshold demand mechanism.

8.3.3 Network charge categories

There are four network charge categories applicable to SACs >100 MWh p.a., as shown in Table 8.1.

Table 8.1: Network charge category and associated threshold demands

Network charge category	Threshold demand
Demand High Voltage (EDHT, WDHT or MIDHT)	400 kW
Demand Large (EDLT, WDLT or MIDLT)	400 kW
Demand Medium (EDMT, WDMT or MIDMT)	120 kW
Demand Small (EDST, WDST or MIDST)	30 kW

The network charge category applicable to a customer will be as nominated by the customer or their retailer. Demand Large, Demand Medium and Demand Small are available to all SACs >100 MWh p.a. These categories are self-selecting, with customers' load characteristics determining the optimum category.

Demand High Voltage is available to customers who have high voltage metering and take supply at high voltage. This category is not available to customers:

- with non-standard or significant connection arrangements
- who have distribution transformer(s) provided by Ergon Energy, or
- who are not metered at high voltage.

8.3.4 DUOS tariffs

DUOS tariffs for all SACs >100 MWh p.a. are based on:

- the averaged allocated connection asset costs
- the averaged allocated shared distribution network costs
- whether the customer is located in the East, West or Mount Isa Zone of Ergon Energy's network.

DUOS tariffs incorporate a fixed charge (\$/day), an actual demand charge (\$/kW/month) and a volume charge (\$/kWh).

The actual demand charge will be applied to the kW amount by which a customer's actual monthly maximum demand is greater than the demand threshold applicable to the customer's network tariff. The following formula is applied:

$$\text{Actual demand charge} = (\text{Metered demand} - \text{Threshold demand}) \times \text{Actual demand charge rate}$$

Where the monthly metered maximum demand is less than the demand threshold, the chargeable demand for that month is set to zero. The threshold demand applicable to each network charge category is shown in Table 8.1.

8.3.5 TUOS tariffs

TUOS charges are calculated and levied for each TNCP. Ergon Energy combines rates in geographical regions in the East and West Zones and passes through TUOS charges to SACs >100 MWh p.a. in these zones, thus providing an appropriate locational signal in each region.

The regional indicator for each TNI is shown in Appendix 1. The indicator aligns with each Network Tariff Code – refer to the suffix of "T". Regional indicators do not apply to the Mount Isa Zone.

TUOS tariffs incorporate a fixed charge (\$/day), capacity charge (\$/kW/month) and volume charge (\$/kWh).

The same DUOS demand threshold calculation mechanism applies for TUOS charges. Where the monthly metered maximum demand is less than the demand threshold, the chargeable demand is set to zero and

no demand charge is payable for that month. Details of the threshold demand for each network charge category are shown in Section 8.3.3 above.

To determine the total TUOS volume charge, the metered consumption must be multiplied by the customer's DLF and then applied to the TUOS \$/kWh rate.

8.3.6 Network Tariff Codes

The Network Tariff Codes that apply to SACs >100 MWh p.a. are provided in Table 8.2.

Table 8.2: Network Tariff Codes, by network charge category for SACs >100 MWh p.a.

Network charge category	Network Tariff Code	Description
Demand High Voltage	EDHTT1	East Demand HV (400 kW) SAC – TUOS Region 1
	EDHTT2	East Demand HV (400 kW) SAC – TUOS Region 2
	EDHTT3	East Demand HV (400 kW) SAC – TUOS Region 3
	WDHTT1	West Demand HV (400 kW) SAC – TUOS Region 1
	WDHTT2	West Demand HV (400 kW) SAC – TUOS Region 2
	WDHTT3	West Demand HV (400 kW) SAC – TUOS Region 3
	MIDHT	Mount Isa Demand HV (400 kW) SAC
Demand Large	EDLTT1	East Demand Large (400 kW) SAC – TUOS Region 1
	EDLTT2	East Demand Large (400 kW) SAC – TUOS Region 2
	EDLTT3	East Demand Large (400 kW) SAC – TUOS Region 3
	WDLTT1	West Demand Large (400 kW) SAC – TUOS Region 1
	WDLTT2	West Demand Large (400 kW) SAC – TUOS Region 2
	WDLTT3	West Demand Large (400 kW) SAC – TUOS Region 3
	MIDLT	Mount Isa Demand Large (400 kW) SAC
Demand Medium	EDMTT1	East Demand Medium (120 kW) SAC – TUOS Region 1
	EDMTT2	East Demand Medium (120 kW) SAC – TUOS Region 2
	EDMTT3	East Demand Medium (120 kW) SAC – TUOS Region 3
	WDMTT1	West Demand Medium (120 kW) SAC – TUOS Region 1
	WDMTT2	West Demand Medium (120 kW) SAC – TUOS Region 2
	WDMTT3	West Demand Medium (120 kW) SAC – TUOS Region 3
	MIDMT	Mount Isa Demand Medium (120 kW) SAC
Demand Small	EDSTT1	East Demand Small (30 kW) SAC – TUOS Region 1
	EDSTT2	East Demand Small (30 kW) SAC – TUOS Region 2
	EDSTT3	East Demand Small (30 kW) SAC – TUOS Region 3
	WDSTT1	West Demand Small (30 kW) SAC – TUOS Region 1
	WDSTT2	West Demand Small (30 kW) SAC – TUOS Region 2
	WDSTT3	West Demand Small (30 kW) SAC – TUOS Region 3
	MIDST	Mount Isa Demand Small (30 kW) SAC

8.4 Standard Asset Customers <100 MWh p.a.

8.4.1 Definition

SACs <100 MWh p.a. are those customers with energy consumption typically less than 100 MWh p.a.

8.4.2 Changes to SAC <100 MWh p.a. tariffs

The changes in 2014–15 include:

- removing the distinction between Volume Small (less than 26 MWh p.a.) and Volume Large (more than 26 MWh p.a. and less than 100 MWh p.a.) network tariffs and introducing segmentation between residential and business customers
- increasing the proportion of revenue recovered from the fixed charge
- adopting an IBT as the default primary tariff
- offering an optional seasonal TOU tariff to market customers.

8.4.3 Network charge categories

There are four primary network charge categories applicable to SACs <100 MWh p.a., as shown in Table 8.3.

Table 8.3: Network charge categories for SACs <100 MWh p.a.

Network charge category
IBT Residential (ERIB, WRIB or MIRIB)
IBT Business (EBIB, WBIB or MIBIB)
TOU Residential (ERTOU, WRTOU or MIRTOU)
TOU Business (EBTOU, WBTOU or MIBTOU)

In addition to these, Ergon Energy has two network charge categories that only apply to SACs <100 MWh p.a. whose primary network charge category is IBT Residential, IBT Business, TOU Residential or TOU Business and where specified connected appliances are controlled by network equipment.⁷

The additional categories that may be nominated by the customer or their retailer are:

- **Volume Night Controlled (EVN, WVN or MIVN)** – applies where specified permanently connected appliances are controlled by network equipment so that supply will be available for a minimum period of eight hours at the absolute discretion of Ergon Energy but usually between the hours of 10:00pm and 7:00am.

⁷ A controlled load pricing option only applies where electricity supply is connected to appliances in a specified manner, separately metered and controlled at a supply point.

For information on the suitability of appliances and other conditions associated with controlled supply, customers should reference the Queensland Government's "Tariff Schedule" for regulated retail electricity prices (or Notified Prices) available at <http://www.dews.qld.gov.au/energy-water-home/electricity/prices/current-prices>. The conditions contained in the Notified Prices will be applied by Ergon Energy to all customers on controlled supply (i.e. customers on a market retail contract as well as customers on a Standard Retail Contract or Standard Large Non-Market Customer Retail Contract).

Switching times will be managed at Ergon Energy's absolute discretion, in accordance with our network requirements.

For customers on a Standard Retail Contract or Standard Large Non-Market Customer Retail Contract, Volume Night Controlled equates to retail Tariff 31 and Volume Controlled equates to retail Tariff 33 in the Notified Prices.

- **Volume Controlled** (EVC, WVC or MIVC) – applies where specified permanently connected appliances are controlled by network equipment so that supply will be available for a minimum period of 18 hours per day during time periods set at the absolute discretion of Ergon Energy.

8.4.4 DUOS tariffs

DUOS tariffs for SACs <100 MWh p.a. are based on:

- the averaged allocated connection asset costs
- the averaged allocated shared distribution network costs
- whether the customer is located in the East, West or Mount Isa Zone of Ergon Energy's network.

Specific information for each network tariff is set out below.

IBT Residential tariff

Ergon Energy's IBT Residential tariff has three inclining blocks.

Respective rates for each of Ergon Energy's consumption blocks apply only to consumption in the block. The consumption blocks for the IBT Residential tariffs are set out in Table 8.4.

Table 8.4: IBT Residential consumption blocks

Block	Daily kWh	Annual equivalent kWh
Block 1	0 – 2.74 kWh	0 – 1,000 kWh p.a.
Block 2	>2.74 – 16.43 kWh	>1,000 kWh p.a. – 6,000 kWh p.a.
Block 3	>16.43 kWh	>6,000 kWh p.a.

Note: For network billing purposes, the IBT is denominated and applied on a daily basis. The annual equivalent is provided for presentation purposes only.

The IBT Residential tariff is the default primary network tariff applying to all residential SAC <100 MWh p.a. customers from 1 July 2014. This means all residential customers currently on the Volume Small and Volume Large network tariffs will transition to the IBT Residential tariff from 1 July 2014.

The IBT Residential tariffs incorporate the following charging parameters:

- a fixed charge (\$/day)
- volume charges (\$/kWh) (specific rates applicable to Block 1, Block 2 and Block 3).

IBT Business tariff

Ergon Energy's IBT Business tariff has three inclining blocks.

Respective rates for each of Ergon Energy's consumption blocks apply only to consumption in the block. The consumption blocks for the IBT Business tariffs are set out in Table 8.5.

Table 8.5: IBT Business consumption blocks

Block	Daily kWh	Annual equivalent kWh
Block 1	0 – 2.74 kWh	0 – 1,000 kWh p.a.
Block 2	>2.74 – 54.76 kWh	>1,000 kWh p.a. – 20,000 kWh p.a.
Block 3	>54.76 kWh	>20,000 kWh p.a.

Note: For network billing purposes, the IBT is denominated and applied on a daily basis. The annual equivalent is provided for presentation purposes only.

The IBT Business tariff is the default primary network tariff applying to all business SAC <100 MWh p.a. customers from 1 July 2014. This means all business customers currently on the Volume Small and Volume Large network tariffs will transition to the IBT Business tariff from 1 July 2014.

The IBT Business tariffs incorporate the following charging parameters:

- a fixed charge (\$/day)
- volume charges (\$/kWh) (specific rates applicable to Block 1, Block 2 and Block 3).

TOU Residential tariff

Ergon Energy's TOU Residential tariff adopts a three-part seasonal TOU tariff structure.

The time periods applying to TOU Residential tariffs are outlined below.

Table 8.6: TOU Residential time periods

Peak	4:30pm to 9:00pm on Summer weekdays
Shoulder	3:00pm to 4:30pm and 9:00pm to 9:30pm on Summer weekdays, and 3:00pm to 9:30pm on Summer weekends
Off-Peak	All other times

Note: Summer is defined as the months of December, January and February.

The TOU Residential tariff is optional for residential SAC <100 MWh p.a. customers from 1 July 2014 who are on a market contract. A customer (or their retailer) must request a tariff change to opt in to these tariffs. This request will be subject to metering considerations.

The TOU Residential tariffs incorporate the following charging parameters:

- a fixed charge (\$/day)
- volume charges (\$/kWh), which will be based on the above TOU structure (i.e. Peak, Shoulder and Off-Peak time periods).

TOU Business tariff

Ergon Energy's TOU Business tariff adopts a three-part seasonal TOU tariff structure.

The time periods applying to TOU Business tariffs are set out below.

Table 8.7: TOU Business time periods

Peak	11:30am to 5:30pm on Summer weekdays
Shoulder	10:00am to 11:30am and 5:30pm to 8:00pm on Summer weekdays
Off-Peak	All other times

Note: Summer is defined as the months of December, January and February.

The TOU Business tariff is optional for business SAC <100 MWh p.a. customers from 1 July 2014 who are on a market contract. A customer (or their retailer) must request a tariff change to opt in to these tariffs. This request will be subject to metering considerations.

The TOU Business tariffs incorporate the following charging parameters:

- a fixed charge (\$/day)
- volume charges (\$/kWh), which will be based on the above TOU structure (i.e. Peak, Shoulder and Off-Peak time periods).

Controlled tariffs

The Volume Night Controlled and Volume Controlled tariffs incorporate the following charging parameters:

- a fixed charge (\$/day)
- a volume charge (\$/kWh).

8.4.5 TUOS tariffs

TUOS charges are calculated and levied for each TNCP. Ergon Energy combines rates in geographical regions in the East and West Zones and passes through TUOS charges to SACs <100 MWh p.a. in these zones, thus providing an appropriate locational signal in each region.

The regional indicator for each TNI is shown in Appendix 1. The indicator aligns with each Network Tariff Code – refer to the suffix of “T”. Regional indicators do not apply to the Mount Isa Zone.

TUOS tariffs for IBT Residential, IBT Business, TOU Residential and TOU Business incorporate a fixed charge (\$/day) and volume charge (\$/kWh). TUOS tariffs for Volume Night Controlled and Volume Controlled incorporate a volume charge (\$/kWh) only.

To determine the total TUOS volume charge, the metered consumption must be multiplied by the customer’s DLF and then applied to the TUOS \$/kWh rate.

8.4.6 Network Tariff Codes

The Network Tariff Codes that apply to SACs <100 MWh p.a. are provided in Table 8.8.

Table 8.8: Network Tariff Codes, by network charge category for SACs <100 MWh p.a.

Network charge category	Network Tariff Code	Description
IBT Residential	ERIBT1	East Residential Inclining Block – TUOS Region 1
	ERIBT2	East Residential Inclining Block – TUOS Region 2
	ERIBT3	East Residential Inclining Block – TUOS Region 3
	WRIBT1	West Residential Inclining Block – TUOS Region 1
	WRIBT2	West Residential Inclining Block – TUOS Region 2
	WRIBT3	West Residential Inclining Block – TUOS Region 3
	MIRIB	Mount Isa Residential Inclining Block
IBT Business	EBIBT1	East Business Inclining Block – TUOS Region 1
	EBIBT2	East Business Inclining Block – TUOS Region 2
	EBIBT3	East Business Inclining Block – TUOS Region 3
	WBIBT1	West Business Inclining Block – TUOS Region 1
	WBIBT2	West Business Inclining Block – TUOS Region 2
	WBIBT3	West Business Inclining Block – TUOS Region 3
	MIBIB	Mount Isa Business Inclining Block
TOU Residential	ERTOUT1	East Residential Time-of-Use – TUOS Region 1
	ERTOUT2	East Residential Time-of-Use – TUOS Region 2
	ERTOUT3	East Residential Time-of-Use – TUOS Region 3
	WRTOUT1	West Residential Time-of-Use – TUOS Region 1
	WRTOUT2	West Residential Time-of-Use – TUOS Region 2
	WRTOUT3	West Residential Time-of-Use – TUOS Region 3
	MIRTOU	Mount Isa Residential Time-of-Use

Network charge category	Network Tariff Code	Description
TOU Business	EBTOUT1	East Business Time-of-Use – TUOS Region 1
	EBTOUT2	East Business Time-of-Use – TUOS Region 2
	EBTOUT3	East Business Time-of-Use – TUOS Region 3
	WBTOUT1	West Business Time-of-Use – TUOS Region 1
	WBTOUT2	West Business Time-of-Use – TUOS Region 2
	WBTOUT3	West Business Time-of-Use – TUOS Region 3
	MIBTOU	Mount Isa Business Time-of-Use
Volume Night Controlled	EVNT1	East Volume Night Controlled (T31) – TUOS Region 1
	EVNT2	East Volume Night Controlled (T31) – TUOS Region 2
	EVNT3	East Volume Night Controlled (T31) – TUOS Region 3
	WVNT1	West Volume Night Controlled (T31) – TUOS Region 1
	WVNT2	West Volume Night Controlled (T31) – TUOS Region 2
	WVNT3	West Volume Night Controlled (T31) – TUOS Region 3
	MIVN	Mount Isa Volume Night Controlled (T31)
Volume Controlled	EVCT1	East Volume Controlled (T33) – TUOS Region 1
	EVCT2	East Volume Controlled (T33) – TUOS Region 2
	EVCT3	East Volume Controlled (T33) – TUOS Region 3
	WVCT1	West Volume Controlled (T33) – TUOS Region 1
	WVCT2	West Volume Controlled (T33) – TUOS Region 2
	WVCT3	West Volume Controlled (T33) – TUOS Region 3
	MIVC	Mount Isa Volume Controlled (T33)

8.5 Standard Asset Customers – unmetered supplies

8.5.1 Definition

SACs – unmetered supplies refers to those customers who take supply where no meter is installed at the connection point. Unmetered supplies should be avoided where possible. However, Ergon Energy may approve an unmetered supply where it meets the requirements set out in Chapter 7 of the NER.

8.5.2 Changes to unmetered supply tariffs

There are no structural changes to unmetered supply tariffs in 2014–15.

8.5.3 Street light installations

The supply of energy to a street lighting connection point is a Standard Control Service and is subject to the same DUOS and TUOS charges applicable to all other unmetered supplies.

Street lighting customers will also be subject to an additional Alternative Control Service daily street lighting charge if Ergon Energy owns and maintains the street lighting assets beyond the customer's connection point (i.e. where Ergon Energy provides ACS – Street Lighting Services). If Ergon Energy does not own and maintain the street lighting assets the Alternative Control Service daily street lighting charge will not apply.

The Alternative Control Service daily street lighting charge which applies in 2014–15 is set out in the “Price List for Alternative Control Services”. Further information on Ergon Energy’s ACS – Street Lighting Services is available in the “Information Guide for Alternative Control Services Pricing”.

8.5.4 Network charge categories

There are three network charge categories applicable to unmetered supplies:

- **Unmetered Supply** (EVU, WVU or MIVU) – applies to small uniform loads approved to be unmetered by Ergon Energy. Includes provision of energy to traffic lights, watchman lights and other types of unmetered public amenities (e.g. illuminated signs, phone boxes and barbecues etc.). It also applies to street light installations where the customer owns and maintains all assets after the customer’s connection point to the network (i.e. where Ergon Energy does not provide ACS – Street Lighting Services).
- **Unmetered Supply with ACS – Street Lighting Services – Minor Street Light** (EVUMI, WVUMI or MIVUMI) – applies to unmetered ‘minor’ street lighting where Ergon Energy owns and maintains all assets after the customer’s connection point to the network.

Ergon Energy’s standard minor street lights are 50, 80 and 125 watt Mercury Vapour and 70 watt High Pressure Sodium vapour lights, and compact fluorescent lights of similar luminescence and include any other non-standard or obsolete street lights that would be replaced with any of the above Ergon Energy standard minor street lights in accordance with Ergon Energy policy.

- **Unmetered Supply with ACS – Street Lighting Services – Major Street Light** (EVUMA, WVUMA or MIVUMA) – applies to unmetered ‘major’ street lighting where Ergon Energy owns and maintains all assets after the customer’s connection point to the network.

Ergon Energy’s standard major street lights are 100, 150, 250 and 400 watt High Pressure Sodium vapour lights and include any other non-standard or obsolete street lights that would be replaced with any of the above Ergon Energy standard major street lights in accordance with Ergon Energy policy.

The network charge category applicable to a customer will be as nominated by the customer or their retailer.

8.5.5 DUOS tariffs

DUOS tariffs for unmetered supplies are based on:

- the average of the connection charges
- the average of the shared distribution network charges
- whether the customer is located in the East, West or Mount Isa Zone of Ergon Energy’s network.

DUOS tariffs incorporate a fixed charge (\$/day) and a volume charge (\$/kWh).

8.5.6 TUOS tariffs

TUOS charges are calculated and levied for each TNCP. Ergon Energy combines rates in geographical regions in the East and West Zones and passes through TUOS charges to SACs in these zones, thus providing an appropriate locational signal in each region.

The regional indicator for each TNI is shown in Appendix 1. The indicator aligns with each Network Tariff Code – refer to the suffix of “T”. Regional indicators do not apply to the Mount Isa Zone.

TUOS tariffs incorporate a volume charge (\$/kWh) only.

To determine the total TUOS volume charge, the consumption must be multiplied by the customer's DLF and then applied to the TUOS \$/kWh rate.

8.5.7 Network Tariff Codes

The Network Tariff Codes that apply to unmetered supplies are shown in Table 8.9 below.

Table 8.9: Network Tariff Codes for unmetered supplies

Network charge category	Network Tariff Code	Description
Unmetered Supply	EVUT1	East Volume Unmetered – TUOS Region 1
	EVUT1MI	East Volume Unmetered – TUOS Region 1 – Minor Street Light
	EVUT1MA	East Volume Unmetered – TUOS Region 1 – Major Street Light
	EVUT2	East Volume Unmetered – TUOS Region 2
	EVUT2MI	East Volume Unmetered – TUOS Region 2 – Minor Street Light
	EVUT2MA	East Volume Unmetered – TUOS Region 2 – Major Street Light
	EVUT3	East Volume Unmetered – TUOS Region 3
	EVUT3MI	East Volume Unmetered – TUOS Region 3 – Minor Street Light
	EVUT3MA	East Volume Unmetered – TUOS Region 3 – Major Street Light
	WVUT1	West Volume Unmetered – TUOS Region 1
	WVUT1MI	West Volume Unmetered – TUOS Region 1 – Minor Street Light
	WVUT1MA	West Volume Unmetered – TUOS Region 1 – Major Street Light
	WVUT2	West Volume Unmetered – TUOS Region 2
	WVUT2MI	West Volume Unmetered – TUOS Region 2 – Minor Street Light
	WVUT2MA	West Volume Unmetered – TUOS Region 2 – Major Street Light
	WVUT3	West Volume Unmetered – TUOS Region 3
	WVUT3MI	West Volume Unmetered – TUOS Region 3 – Minor Street Light
	WVUT3MA	West Volume Unmetered – TUOS Region 3 – Major Street Light
	MIVU	Mount Isa Volume Unmetered
	MIVUMI	Mount Isa Volume Unmetered – Minor Street Light
	MIVUMA	Mount Isa Volume Unmetered – Major Street Light

8.6 Embedded Generators

8.6.1 Changes to EG tariffs

There have been no structural changes for EG tariffs in 2014–15.

8.6.2 DUOS tariffs

For those EGs that generate into the distribution system a fixed charge (\$/day) applies.

For those EGs that generate into, as well as take load from the distribution system, a fixed charge (\$/day) applies, and DUOS tariffs for load taken from the distribution system will be allocated as per the appropriate network user group (i.e. ICC, CAC or SAC).

All EG DUOS tariffs are based on:

- whether the customer is subject to the pre or post 30 June 2010 arrangements for Large Customer Connections
- whether the customer is located in the East, West or Mount Isa Zone of Ergon Energy's network.

8.6.3 TUOS tariffs

For those EGs that only generate into the distribution system, TUOS tariffs do not apply to generated energy.

For those EGs that generate into, as well as take load from the distribution system:

- TUOS tariffs do not apply to generated energy
- TUOS tariffs for load taken from the distribution system will be allocated as per the appropriate network user group (i.e. ICC, CAC or SAC).

8.7 Queensland Government Solar Bonus Scheme

The Queensland Government Solar Bonus Scheme is a program that pays eligible customers for the surplus electricity generated from solar PV systems that is exported to the Queensland electricity grid. For information on the Solar Bonus Scheme, please refer to:

<http://www.dews.qld.gov.au/energy-water-home/electricity/solar-bonus-scheme>.

8.7.1 Network Tariff Codes

The Network Tariff Codes that apply to the Queensland Government Solar Bonus Scheme are detailed in Table 8.10 below.

Table 8.10: Network Tariff Codes for the Solar Bonus Scheme

Network Tariff Code	Description
NVG0	Net Volume Generation 0
NVG1	Net Volume Generation 1
NGV2	Net Volume Generation 2
GVG0	Gross Volume Generation 0

Where:

- NVG0 is applicable to any net metering installation that does not meet the Solar Bonus Scheme requirements
- NVG1 applies to any net metering installation that meets the Solar Bonus Scheme requirements for the 44 c/kWh feed-in tariff rate
- NVG2 applies to any net metering installation that meets the Solar Bonus Scheme requirements (other than the 44 c/kWh feed-in tariff rate)⁸
- GVG0 is applicable to any gross metering installation.

⁸ This Network Tariff Code was previously associated with the distributor funded 8 c/kWh feed-in tariff rate, which ended on 30 June 2014. From 1 July 2014, the 8 c/kWh feed-in tariff in regional Queensland will be replaced by a mandated feed-in tariff set by the QCA. The feed-in tariff will be paid by the electricity retailer. Consequently, the rate associated with this Network Tariff Code will be 0 c/kWh.

Appendix 1: TUOS regional indicators

TUOS regional indicators for 2014–15		
Standard Asset Customers		
TNCP	TNI	TUOS region
Alligator Creek	QALC	T2
Louisa Creek	QALH	T2
Alan Sherriff (11)	QASF	T2
Biloela	QBIL	T1
Bulli Creek	QBLK	T1
Bowen North	QBNN	T2
Blackwater (132)	QBWH	T1
Blackwater (66)	QBWL	T1
Calliope River	QCAR	T1
Columboola	QCBL	T1
Cardwell	QCDW	T3
Chinchilla	QCHA	T1
Clare	QCLR	T2
Cairns (Cairns City) (132)	QCNS	T3
Collinsville	QCOL	T2
Cairns (22)	QCRN	T3
Dan Gleeson	QDGL	T2
Dysart	QDYS	T1
Egans Hill	QEGN	T1
El Arish	QELA	T3
Kemmis	QEMS	T2
Edmonton	QEMT	T3
Alan Sherriff (Garbutt)	QGAR	T2
Gin Gin	QGNG	T1
Gladstone South	QGST	T1
Innisfail	QINF	T3
Ingham	QING	T3
Kamerunga	QKAM	T3
King Creek	QKCK	T2
Lilyvale (Clermont) (132)	QLCM	T1
Lilyvale (66)	QLIL	T1
Mackay	QMKA	T2
Moura	QMRA	T1
Middle Ridge	QMRG	T1

TUOS regional indicators for 2014–15		
Standard Asset Customers		
TNCP	TNI	TUOS region
Broadlea	QMRH	T1
Moranbah (11)	QMRL	T1
Moranbah (66)	QMRN	T1
Nebo	QNEB	T2
Newlands (Mackay & Northern)	QNLD	T2
Oakey Town	QOKT	T1
Pandoin	QPAL	T1
Pioneer Valley	QPIV	T2
Proserpine (Mackay)	QPRO	T2
Rockhampton	QROC	T1
Ross (Kidston & Millchester)	QROS	T2
Stony Creek	QSYS	T2
Teebar Creek	QTBC	T1
Tangkam	QTKM	T1
Tully	QTLL	T3
Tarong (66)	QTRL	T1
Turkinje (Port Douglas) (132)	QTUH	T3
Turkinje (66)	QTUL	T3
Townsville East	QTVE	T2
Townsville South	QTVS	T2
Woolooga	QWLN	T1
Woree	QWRE	T3
Boat Creek	QYAE	T1

Appendix 2: 2014–15 Network Tariff Codes and rates

2014-15 Rates										
TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – LARGE (>100 MWh per annum) GST Exclusive										
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)			Transmission Use Of System (TUOS)		
					Fixed Charge	Actual Demand Charge	Volume Charge	Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
Demand High Voltage	EDHTT1	400	1.034	GEHL	\$341.824	\$20.965	\$0.00553	\$17.298	\$0.959	\$0.01110
	EDHTT2	400	1.034	GEHL	\$341.824	\$20.965	\$0.00553	\$33.843	\$2.275	\$0.01330
	EDHTT3	400	1.034	GEHL	\$341.824	\$20.965	\$0.00553	\$63.438	\$4.532	\$0.01668
	WDHTT1	400	1.127	GWHL	\$800.148	\$50.000	\$0.01526	\$17.298	\$0.959	\$0.01110
	WDHTT2	400	1.127	GWHL	\$800.148	\$50.000	\$0.01526	\$33.843	\$2.275	\$0.01330
	WDHTT3	400	1.127	GWHL	\$800.148	\$50.000	\$0.01526	\$63.438	\$4.532	\$0.01668
	MIDHT	400	1.035	GMHL	POA	POA	POA	\$8.128	\$0.449	\$0.00047
Demand Large	EDLTT1	400	1.094	GELL	\$419.276	\$28.780	\$0.00553	\$17.298	\$0.959	\$0.01110
	EDLTT2	400	1.094	GELL	\$419.276	\$28.780	\$0.00553	\$33.843	\$2.275	\$0.01330
	EDLTT3	400	1.094	GELL	\$419.276	\$28.780	\$0.00553	\$63.438	\$4.532	\$0.01668
	WDLTT1	400	1.184	GWLL	\$1,254.653	\$90.000	\$0.00960	\$17.298	\$0.959	\$0.01110
	WDLTT2	400	1.184	GWLL	\$1,254.653	\$90.000	\$0.00960	\$33.843	\$2.275	\$0.01330
	WDLTT3	400	1.184	GWLL	\$1,254.653	\$90.000	\$0.00960	\$63.438	\$4.532	\$0.01668
	MIDLT	400	1.060	GMLL	\$262.385	\$15.000	\$0.00493	\$8.128	\$0.449	\$0.00047
Demand Medium	EDMTT1	120	1.094	GELL	\$140.448	\$30.076	\$0.00553	\$8.476	\$0.959	\$0.01110
	EDMTT2	120	1.094	GELL	\$140.448	\$30.076	\$0.00553	\$12.915	\$2.275	\$0.01330
	EDMTT3	120	1.094	GELL	\$140.448	\$30.076	\$0.00553	\$21.747	\$4.532	\$0.01668
	WDMTT1	120	1.184	GWLL	\$394.216	\$92.000	\$0.01309	\$8.476	\$0.959	\$0.01110
	WDMTT2	120	1.184	GWLL	\$394.216	\$92.000	\$0.01309	\$12.915	\$2.275	\$0.01330
	WDMTT3	120	1.184	GWLL	\$394.216	\$92.000	\$0.01309	\$21.747	\$4.532	\$0.01668
	MIDMT	120	1.060	GMLL	\$92.106	\$18.284	\$0.00493	\$3.997	\$0.449	\$0.00047

2014-15 Rates										
TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – LARGE (>100 MWh per annum) GST Exclusive										
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)			Transmission Use Of System (TUOS)		
					Fixed Charge	Actual Demand Charge	Volume Charge	Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
Demand Small	EDSTT1	30	1.094	GELL	\$38.728	\$33.630	\$0.00553	\$5.640	\$0.959	\$0.01110
	EDSTT2	30	1.094	GELL	\$38.728	\$33.630	\$0.00553	\$6.188	\$2.275	\$0.01330
	EDSTT3	30	1.094	GELL	\$38.728	\$33.630	\$0.00553	\$8.347	\$4.532	\$0.01668
	WDSTT1	30	1.184	GWLL	\$107.011	\$95.846	\$0.01526	\$5.640	\$0.959	\$0.01110
	WDSTT2	30	1.184	GWLL	\$107.011	\$95.846	\$0.01526	\$6.188	\$2.275	\$0.01330
	WDSTT3	30	1.184	GWLL	\$107.011	\$95.846	\$0.01526	\$8.347	\$4.532	\$0.01668
	MIDST	30	1.060	GMLL	\$25.900	\$21.664	\$0.00493	\$2.670	\$0.449	\$0.00047

2014-15 Rates											
TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – SMALL (<100 MWh per annum) INCLINING BLOCK TARIFF (IBT) GST Exclusive											
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)				Transmission Use Of System (TUOS)		
					Fixed Charge	Volume Charge *			Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code		Block 1	Block 2	Block 3			
					dollars per day	dollars per kilowatt hour	dollars per kilowatt hour	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
IBT Residential	ERIBT1		1.094	GELL	\$1.525	\$0.00000	\$0.15314	\$0.16314	\$0.121		\$0.01110
	ERIBT2		1.094	GELL	\$1.525	\$0.00000	\$0.15314	\$0.16314	\$0.218		\$0.01330
	ERIBT3		1.094	GELL	\$1.525	\$0.00000	\$0.15314	\$0.16314	\$0.297		\$0.01668
	WRIBT1		1.184	GWLL	\$2.325	\$0.07000	\$0.44971	\$0.45971	\$0.121		\$0.01110
	WRIBT2		1.184	GWLL	\$2.325	\$0.07000	\$0.44971	\$0.45971	\$0.218		\$0.01330
	WRIBT3		1.184	GWLL	\$2.325	\$0.07000	\$0.44971	\$0.45971	\$0.297		\$0.01668
	MIRIB		1.060	GMLL	\$1.545	\$0.00000	\$0.07771	\$0.08771	\$0.107		\$0.00047
IBT Business	EBIBT1		1.094	GELL	\$1.525	\$0.00000	\$0.15383	\$0.16383	\$0.121		\$0.01110
	EBIBT2		1.094	GELL	\$1.525	\$0.00000	\$0.15383	\$0.16383	\$0.218		\$0.01330
	EBIBT3		1.094	GELL	\$1.525	\$0.00000	\$0.15383	\$0.16383	\$0.297		\$0.01668
	WBIBT1		1.184	GWLL	\$2.295	\$0.07000	\$0.45571	\$0.46571	\$0.121		\$0.01110
	WBIBT2		1.184	GWLL	\$2.295	\$0.07000	\$0.45571	\$0.46571	\$0.218		\$0.01330
	WBIBT3		1.184	GWLL	\$2.295	\$0.07000	\$0.45571	\$0.46571	\$0.297		\$0.01668
	MIBIB		1.060	GMLL	\$1.545	\$0.00000	\$0.08138	\$0.09978	\$0.107		\$0.00047

* The consumption blocks applying to the volume charge of the IBT Residential and IBT Business tariffs are set out in Section 8.4.4.

2014-15 Rates TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – SMALL (<100 MWh per annum) TIME-OF-USE (TOU) GST Exclusive											
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)				Transmission Use Of System (TUOS)		
					Fixed Charge	Volume Charge *					
		kilowatts	Value	Code		dollars per day	dollars per kilowatt hour	dollars per kilowatt hour	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month
TOU Residential	ERTOUT1		1.094	GELL	\$1.525	\$0.55194	\$0.26664	\$0.09568	\$0.121		\$0.01110
	ERTOUT2		1.094	GELL	\$1.525	\$0.55194	\$0.26664	\$0.09568	\$0.218		\$0.01330
	ERTOUT3		1.094	GELL	\$1.525	\$0.55194	\$0.26664	\$0.09568	\$0.297		\$0.01668
	WRTOUT1		1.184	GWLL	\$2.325	\$1.51214	\$0.96023	\$0.31361	\$0.121		\$0.01110
	WRTOUT2		1.184	GWLL	\$2.325	\$1.51214	\$0.96023	\$0.31361	\$0.218		\$0.01330
	WRTOUT3		1.184	GWLL	\$2.325	\$1.51214	\$0.96023	\$0.31361	\$0.297		\$0.01668
	MIRTOU		1.060	GMLL	\$1.545	\$0.55194	\$0.35049	\$0.02532	\$0.107		\$0.00047
TOU Business	EBTOUT1		1.094	GELL	\$1.525	\$0.41395	\$0.30663	\$0.12364	\$0.121		\$0.01110
	EBTOUT2		1.094	GELL	\$1.525	\$0.41395	\$0.30663	\$0.12364	\$0.218		\$0.01330
	EBTOUT3		1.094	GELL	\$1.525	\$0.41395	\$0.30663	\$0.12364	\$0.297		\$0.01668
	WBTOUT1		1.184	GWLL	\$2.295	\$1.13411	\$1.10426	\$0.35443	\$0.121		\$0.01110
	WBTOUT2		1.184	GWLL	\$2.295	\$1.13411	\$1.10426	\$0.35443	\$0.218		\$0.01330
	WBTOUT3		1.184	GWLL	\$2.295	\$1.13411	\$1.10426	\$0.35443	\$0.297		\$0.01668
	MIBTOU		1.060	GMLL	\$1.545	\$0.41395	\$0.40306	\$0.04885	\$0.107		\$0.00047

* The Peak, Shoulder and Off-Peak time periods applying to the volume charge of the TOU Residential and TOU Business tariffs are set out in Section 8.4.4.

2014-15 Rates TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – SMALL (<100 MWh per annum) CONTROLLED LOAD GST Exclusive										
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)			Transmission Use Of System (TUOS)		
					Fixed Charge	Actual Demand Charge	Volume Charge	Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
Volume Night Controlled	EVNT1		1.094	GELL	\$0.117		\$0.00360			\$0.01110
	EVNT2		1.094	GELL	\$0.117		\$0.00360			\$0.01330
	EVNT3		1.094	GELL	\$0.117		\$0.00360			\$0.01668
	WVNT1		1.184	GWLL	\$0.138		\$0.01139			\$0.01110
	WVNT2		1.184	GWLL	\$0.138		\$0.01139			\$0.01330
	WVNT3		1.184	GWLL	\$0.138		\$0.01139			\$0.01668
	MIVN		1.060	GMLL	\$0.148		\$0.00335			\$0.00047
Volume Controlled	EVCT1		1.094	GELL	\$0.117		\$0.03262			\$0.01110
	EVCT2		1.094	GELL	\$0.117		\$0.03262			\$0.01330
	EVCT3		1.094	GELL	\$0.117		\$0.03262			\$0.01668
	WVCT1		1.184	GWLL	\$0.138		\$0.09154			\$0.01110
	WVCT2		1.184	GWLL	\$0.138		\$0.09154			\$0.01330
	WVCT3		1.184	GWLL	\$0.138		\$0.09154			\$0.01668
	MIVC		1.060	GMLL	\$0.148		\$0.01253			\$0.00047

TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – UNMETERED GST Exclusive										
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)			Transmission Use Of System (TUOS)		
					Fixed Charge	Actual Demand Charge	Volume Charge	Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
Unmetered Supply	EVUT1		1.094	GELL	\$0.006		\$0.19544			\$0.01110
	EVUT1MI		1.094	GELL	\$0.006		\$0.19544			\$0.01110
	EVUT1MA		1.094	GELL	\$0.006		\$0.19544			\$0.01110
	EVUT2		1.094	GELL	\$0.006		\$0.19544			\$0.01330
	EVUT2MI		1.094	GELL	\$0.006		\$0.19544			\$0.01330
	EVUT2MA		1.094	GELL	\$0.006		\$0.19544			\$0.01330
	EVUT3		1.094	GELL	\$0.006		\$0.19544			\$0.01668
	EVUT3MI		1.094	GELL	\$0.006		\$0.19544			\$0.01668
	EVUT3MA		1.094	GELL	\$0.006		\$0.19544			\$0.01668
	WVUT1		1.184	GWLL	\$0.276		\$0.16070			\$0.01110
	WVUT1MI		1.184	GWLL	\$0.276		\$0.16070			\$0.01110
	WVUT1MA		1.184	GWLL	\$0.276		\$0.16070			\$0.01110
	WVUT2		1.184	GWLL	\$0.276		\$0.16070			\$0.01330
	WVUT2MI		1.184	GWLL	\$0.276		\$0.16070			\$0.01330
	WVUT2MA		1.184	GWLL	\$0.276		\$0.16070			\$0.01330
	WVUT3		1.184	GWLL	\$0.276		\$0.16070			\$0.01668
	WVUT3MI		1.184	GWLL	\$0.276		\$0.16070			\$0.01668
	WVUT3MA		1.184	GWLL	\$0.276		\$0.16070			\$0.01668
	MIVU		1.060	GMLL	\$0.229		\$0.03330			\$0.00047
	MIVUMI		1.060	GMLL	\$0.229		\$0.03330			\$0.00047
	MIVUMA		1.060	GMLL	\$0.229		\$0.03330			\$0.00047

2014-15 Rates										
TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – LARGE (>100 MWh per annum)										
GST Inclusive										
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)			Transmission Use Of System (TUOS)		
					Fixed Charge	Actual Demand Charge	Volume Charge	Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
Demand High Voltage	EDHTT1	400	1.034	GEHL	\$376.007	\$23.062	\$0.00608	\$19.028	\$1.055	\$0.01221
	EDHTT2	400	1.034	GEHL	\$376.007	\$23.062	\$0.00608	\$37.228	\$2.503	\$0.01463
	EDHTT3	400	1.034	GEHL	\$376.007	\$23.062	\$0.00608	\$69.782	\$4.985	\$0.01835
	WDHTT1	400	1.127	GWHL	\$880.163	\$55.000	\$0.01679	\$19.028	\$1.055	\$0.01221
	WDHTT2	400	1.127	GWHL	\$880.163	\$55.000	\$0.01679	\$37.228	\$2.503	\$0.01463
	WDHTT3	400	1.127	GWHL	\$880.163	\$55.000	\$0.01679	\$69.782	\$4.985	\$0.01835
	MIDHT	400	1.035	GMHL	POA	POA	POA	\$8.941	\$0.494	\$0.00052
Demand Large	EDLTT1	400	1.094	GELL	\$461.203	\$31.658	\$0.00608	\$19.028	\$1.055	\$0.01221
	EDLTT2	400	1.094	GELL	\$461.203	\$31.658	\$0.00608	\$37.228	\$2.503	\$0.01463
	EDLTT3	400	1.094	GELL	\$461.203	\$31.658	\$0.00608	\$69.782	\$4.985	\$0.01835
	WDLTT1	400	1.184	GWLL	\$1,380.118	\$99.000	\$0.01056	\$19.028	\$1.055	\$0.01221
	WDLTT2	400	1.184	GWLL	\$1,380.118	\$99.000	\$0.01056	\$37.228	\$2.503	\$0.01463
	WDLTT3	400	1.184	GWLL	\$1,380.118	\$99.000	\$0.01056	\$69.782	\$4.985	\$0.01835
	MIDLT	400	1.060	GMLL	\$288.623	\$16.500	\$0.00542	\$8.941	\$0.494	\$0.00052
Demand Medium	EDMTT1	120	1.094	GELL	\$154.493	\$33.084	\$0.00608	\$9.323	\$1.055	\$0.01221
	EDMTT2	120	1.094	GELL	\$154.493	\$33.084	\$0.00608	\$14.207	\$2.503	\$0.01463
	EDMTT3	120	1.094	GELL	\$154.493	\$33.084	\$0.00608	\$23.922	\$4.985	\$0.01835
	WDMTT1	120	1.184	GWLL	\$433.638	\$101.200	\$0.01440	\$9.323	\$1.055	\$0.01221
	WDMTT2	120	1.184	GWLL	\$433.638	\$101.200	\$0.01440	\$14.207	\$2.503	\$0.01463
	WDMTT3	120	1.184	GWLL	\$433.638	\$101.200	\$0.01440	\$23.922	\$4.985	\$0.01835
	MIDMT	120	1.060	GMLL	\$101.316	\$20.112	\$0.00542	\$4.397	\$0.494	\$0.00052

2014-15 Rates										
TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – LARGE (>100 MWh per annum)										
GST Inclusive										
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)			Transmission Use Of System (TUOS)		
					Fixed Charge	Actual Demand Charge	Volume Charge	Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
Demand Small	EDSTT1	30	1.094	GELL	\$42.600	\$36.993	\$0.00608	\$6.204	\$1.055	\$0.01221
	EDSTT2	30	1.094	GELL	\$42.600	\$36.993	\$0.00608	\$6.807	\$2.503	\$0.01463
	EDSTT3	30	1.094	GELL	\$42.600	\$36.993	\$0.00608	\$9.182	\$4.985	\$0.01835
	WDSTT1	30	1.184	GWLL	\$117.712	\$105.430	\$0.01679	\$6.204	\$1.055	\$0.01221
	WDSTT2	30	1.184	GWLL	\$117.712	\$105.430	\$0.01679	\$6.807	\$2.503	\$0.01463
	WDSTT3	30	1.184	GWLL	\$117.712	\$105.430	\$0.01679	\$9.182	\$4.985	\$0.01835
	MIDST	30	1.060	GMLL	\$28.490	\$23.831	\$0.00542	\$2.937	\$0.494	\$0.00052

2014-15 Rates TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – SMALL (<100 MWh per annum) INCLINING BLOCK TARIFF (IBT) GST Inclusive											
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)				Transmission Use Of System (TUOS)		
					Fixed Charge	Volume Charge *			Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code		Block 1	Block 2	Block 3			
					dollars per day	dollars per kilowatt hour	dollars per kilowatt hour	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
IBT Residential	ERIBT1		1.094	GELL	\$1.67715	\$0.00000	\$0.16845	\$0.17945	\$0.133		\$0.01221
	ERIBT2		1.094	GELL	\$1.67715	\$0.00000	\$0.16845	\$0.17945	\$0.240		\$0.01463
	ERIBT3		1.094	GELL	\$1.67715	\$0.00000	\$0.16845	\$0.17945	\$0.327		\$0.01835
	WRIBT1		1.184	GWLL	\$2.55715	\$0.07700	\$0.49469	\$0.50569	\$0.133		\$0.01221
	WRIBT2		1.184	GWLL	\$2.55715	\$0.07700	\$0.49469	\$0.50569	\$0.240		\$0.01463
	WRIBT3		1.184	GWLL	\$2.55715	\$0.07700	\$0.49469	\$0.50569	\$0.327		\$0.01835
	MIRIB		1.060	GMLL	\$1.69915	\$0.00000	\$0.08548	\$0.09648	\$0.118		\$0.00052
IBT Business	EBIBT1		1.094	GELL	\$1.67715	\$0.00000	\$0.16922	\$0.18022	\$0.133		\$0.01221
	EBIBT2		1.094	GELL	\$1.67715	\$0.00000	\$0.16922	\$0.18022	\$0.240		\$0.01463
	EBIBT3		1.094	GELL	\$1.67715	\$0.00000	\$0.16922	\$0.18022	\$0.327		\$0.01835
	WBIBT1		1.184	GWLL	\$2.52462	\$0.07700	\$0.50128	\$0.51228	\$0.133		\$0.01221
	WBIBT2		1.184	GWLL	\$2.52462	\$0.07700	\$0.50128	\$0.51228	\$0.240		\$0.01463
	WBIBT3		1.184	GWLL	\$2.52462	\$0.07700	\$0.50128	\$0.51228	\$0.327		\$0.01835
	MIBIB		1.060	GMLL	\$1.69915	\$0.00000	\$0.08952	\$0.10976	\$0.118		\$0.00052

* The consumption blocks applying to the volume charge of the IBT Residential and IBT Business tariffs are set out in Section 8.4.4.

2014-15 Rates TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – SMALL (<100 MWh per annum) TIME-OF-USE (TOU) GST Inclusive											
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)				Transmission Use Of System (TUOS)		
					Fixed Charge	Volume Charge *			Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code		Peak	Shoulder	Off-Peak			
					dollars per day	dollars per kilowatt hour	dollars per kilowatt hour	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
TOU Residential	ERTOUT1		1.094	GELL	\$1.67715	\$0.60713	\$0.29330	\$0.10524	\$0.133		\$0.01221
	ERTOUT2		1.094	GELL	\$1.67715	\$0.60713	\$0.29330	\$0.10524	\$0.240		\$0.01463
	ERTOUT3		1.094	GELL	\$1.67715	\$0.60713	\$0.29330	\$0.10524	\$0.327		\$0.01835
	WRTOUT1		1.184	GWLL	\$2.55715	\$1.66336	\$1.05625	\$0.34497	\$0.133		\$0.01221
	WRTOUT2		1.184	GWLL	\$2.55715	\$1.66336	\$1.05625	\$0.34497	\$0.240		\$0.01463
	WRTOUT3		1.184	GWLL	\$2.55715	\$1.66336	\$1.05625	\$0.34497	\$0.327		\$0.01835
	MIRTOU		1.060	GMLL	\$1.69915	\$0.60713	\$0.38553	\$0.02785	\$0.118		\$0.00052
TOU Business	EBTOUT1		1.094	GELL	\$1.67715	\$0.45535	\$0.33729	\$0.13601	\$0.133		\$0.01221
	EBTOUT2		1.094	GELL	\$1.67715	\$0.45535	\$0.33729	\$0.13601	\$0.240		\$0.01463
	EBTOUT3		1.094	GELL	\$1.67715	\$0.45535	\$0.33729	\$0.13601	\$0.327		\$0.01835
	WBTOUT1		1.184	GWLL	\$2.52462	\$1.24752	\$1.21469	\$0.38987	\$0.133		\$0.01221
	WBTOUT2		1.184	GWLL	\$2.52462	\$1.24752	\$1.21469	\$0.38987	\$0.240		\$0.01463
	WBTOUT3		1.184	GWLL	\$2.52462	\$1.24752	\$1.21469	\$0.38987	\$0.327		\$0.01835
	MIBTOU		1.060	GMLL	\$1.69915	\$0.45535	\$0.44336	\$0.05373	\$0.118		\$0.00052

* The Peak, Shoulder and Off-Peak time periods applying to the volume charge of the TOU Residential and TOU Business tariffs are set out in Section 8.4.4.

2014-15 Rates										
TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – SMALL (<100 MWh per annum)										
CONTROLLED LOAD										
GST Inclusive										
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)			Transmission Use Of System (TUOS)		
					Fixed Charge	Actual Demand Charge	Volume Charge	Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
Volume Night Controlled	EVNT1		1.203	GELL	\$0.129		\$0.00396			\$0.01221
	EVNT2		1.203	GELL	\$0.129		\$0.00396			\$0.01463
	EVNT3		1.203	GELL	\$0.129		\$0.00396			\$0.01835
	WVNT1		1.302	GWLL	\$0.152		\$0.01253			\$0.01221
	WVNT2		1.302	GWLL	\$0.152		\$0.01253			\$0.01463
	WVNT3		1.302	GWLL	\$0.152		\$0.01253			\$0.01835
	MIVN		1.166	GMLL	\$0.163		\$0.00369			\$0.00052
Volume Controlled	EVCT1		1.203	GELL	\$0.129		\$0.03588			\$0.01221
	EVCT2		1.203	GELL	\$0.129		\$0.03588			\$0.01463
	EVCT3		1.203	GELL	\$0.129		\$0.03588			\$0.01835
	WVCT1		1.302	GWLL	\$0.152		\$0.10069			\$0.01221
	WVCT2		1.302	GWLL	\$0.152		\$0.10069			\$0.01463
	WVCT3		1.302	GWLL	\$0.152		\$0.10069			\$0.01835
	MIVC		1.166	GMLL	\$0.163		\$0.01378			\$0.00052

2014-15 Rates										
TARIFFS APPLYING TO STANDARD ASSET CUSTOMERS – UNMETERED										
GST Inclusive										
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)			Transmission Use Of System (TUOS)		
					Fixed Charge	Actual Demand Charge	Volume Charge	Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
Unmetered Supply	EVUT1		1.094	GELL	\$0.007		\$0.21498			\$0.01221
	EVUT1MI		1.094	GELL	\$0.007		\$0.21498			\$0.01221
	EVUT1MA		1.094	GELL	\$0.007		\$0.21498			\$0.01221
	EVUT2		1.094	GELL	\$0.007		\$0.21498			\$0.01463
	EVUT2MI		1.094	GELL	\$0.007		\$0.21498			\$0.01463
	EVUT2MA		1.094	GELL	\$0.007		\$0.21498			\$0.01463
	EVUT3		1.094	GELL	\$0.007		\$0.21498			\$0.01835
	EVUT3MI		1.094	GELL	\$0.007		\$0.21498			\$0.01835
	EVUT3MA		1.094	GELL	\$0.007		\$0.21498			\$0.01835
	WVUT1		1.184	GWLL	\$0.304		\$0.17677			\$0.01221
	WVUT1MI		1.184	GWLL	\$0.304		\$0.17677			\$0.01221
	WVUT1MA		1.184	GWLL	\$0.304		\$0.17677			\$0.01221
	WVUT2		1.184	GWLL	\$0.304		\$0.17677			\$0.01463
	WVUT2MI		1.184	GWLL	\$0.304		\$0.17677			\$0.01463
	WVUT2MA		1.184	GWLL	\$0.304		\$0.17677			\$0.01463
	WVUT3		1.184	GWLL	\$0.304		\$0.17677			\$0.01835
	WVUT3MI		1.184	GWLL	\$0.304		\$0.17677			\$0.01835
	WVUT3MA		1.184	GWLL	\$0.304		\$0.17677			\$0.01835
	MIVU		1.060	GMLL	\$0.252		\$0.03663			\$0.00052
	MIVUMI		1.060	GMLL	\$0.252		\$0.03663			\$0.00052
	MIVUMA		1.060	GMLL	\$0.252		\$0.03663			\$0.00052

QUEENSLAND GOVERNMENT SOLAR BONUS SCHEME FOR STANDARD ASSET CUSTOMERS - SMALL (<100 MWh per annum) GST Inclusive *										
Tariff Description	Network Tariff Code	Threshold Above Which Demand Charge Applies	Default Distribution Loss Factor		Distribution Use Of System (DUOS)			Transmission Use Of System (TUOS)		
					Fixed Charge	Actual Demand Charge	Volume Charge	Fixed Charge	Capacity Charge	Volume Charge
		kilowatts	Value	Code	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour	dollars per day	dollars per kilowatt per month	dollars per kilowatt hour
Queensland Government Solar Bonus Scheme	NVG0						\$0.0000			
	NVG1						-\$0.4400			
	NVG2						\$0.0000 **			
	GVG0						\$0.0000			

* The 44 c/kWh feed-in tariff associated with the Solar Bonus Scheme includes GST (if any). As such, no amount will be added to the credit in respect of any GST payable.

** The 8 c/kWh feed-in tariff rate previously associated with this Network Tariff Code ended on 30 June 2014.

Appendix 3: Ergon Energy's IBT network tariff calculation methodology

The IBT is structured with a fixed rate per day and three energy consumption blocks, each with a different volume (energy) rate applicable.

Block and fixed rates are different between the East, West and Mount Isa Zones and between residential and business customers. Block sizes are also different between residential and business customers.

The IBT is denominated and applied on a daily basis. However, it may be described in the context of an annual basis for network tariff consultation and presentation purposes. For example, the IBT Residential consumption blocks are:

Block	Daily kWh	Annual equivalent kWh
Block 1	0 – 2.74 kWh	0 – 1,000 kWh p.a.
Block 2	>2.74 – 16.43 kWh	>1,000 kWh p.a. – 6,000 kWh p.a.
Block 3	>16.43 kWh	>6,000 kWh p.a.

Daily denomination ensures IBT billing is equitably applied for any meter reading period (including NMIs where a customer move-out/move-in occurs), based on an accumulated total of consumption divided by the number of days in the reading period.

The IBT network tariff calculation methodology is as follows:

- a meter read is taken, with total consumption for the number of days within that meter reading period
- for IBT network billing purposes, the energy (kWh) assigned to each block is prorated back to a daily equivalent for that meter reading period
- the bill is then calculated with the component parts being a daily Fixed Charge, Volume Charge Block 1, Volume Charge Block 2 and Volume Charge Block 3
- daily calculations are then converted back to the total network charge by multiplication by the number of days in the meter reading period
- in relation to TUOS –
 - the TUOS volume component will effectively remain as a flat rate for all IBT tariffs (i.e. same TUOS rate applied in each tariff block or meter reading period)
 - the DLF will be applied to total metered consumption for TUOS.

The IBT bill is calculated as follows:

Component	Calculation
Fixed Charge	Number of days in the period <i>multiplied by</i> fixed charge rate.
Volume Charge Block 1	If equivalent daily consumption is less than the Block 1 daily allowance then: ▪ equivalent daily consumption <i>multiplied by</i> Block 1 rate ▪ multiplied by days in meter reading period ▪ no further calculations required. Or, if equivalent daily consumption exceeds the Block 1 daily allowance then: ▪ Block 1 volume charge calculation applied ▪ proceed to Volume Charge Block 2 calculation.
Volume Charge Block 2	For consumption within the Block 2 allowance then: ▪ equivalent daily consumption <i>less</i> the Block 1 daily allowance <i>multiplied by</i>

Component	Calculation
	Block 2 rate (up to the max block 2 threshold) multiplied by days in meter reading period.
Volume Charge Block 3	For any consumption above the aggregate of Block 1 and Block 2 daily allowance then: - remaining consumption above the Block 1 and Block 2 daily allowance <i>multiplied by</i> the Block 3 rate - multiplied by days in meter reading period.

Example: A residential customer is read quarterly. On this occasion, the customer is read at 90 days with the start read being 123,400 kWh and the end read as 125,200 kWh. This equates to a consumption of 1,800 kWh for that quarter's read.

Currently, the GST exclusive rates for each DUOS component of the IBT Residential East tariff are as follows:

Tariff	Fixed	Block 1 Rate	Block 2 Rate	Block 3 Rate
ERIBT1	\$1.525	\$0.0000	\$0.1531	\$0.1631

Calculation:

Equivalent Daily Consumption = consumption *divided by* the number of days in the read = $1,800/90 = 20.00$.
Fixed Charge = $90 \times \$1.525 = \137.2500
Volume Charge Block 1 = $2.74 \times \$0.00 \times 90 = \0
Volume Block 2 = $(16.43 - 2.74) \times \$0.1531 \times 90 = \188.6345
Volume Block 3 = $(20 - 2.74 - 13.69) \times \$0.1631 \times 90 = \$52.4040$
Quarterly Bill = \$378.29

The second quarter for this customer shows a total consumption of 200 kWh over 88 days for the period.

Calculation:

Equivalent Daily Consumption = consumption *divided by* the number of days in the read = $200/88 = 2.273$
Fixed Charge = $88 \times \$1.525 = \134.2000
Volume Block 1 = $2.273 \times \$0.00 = \0
Volume Block 2 = $0 \times \$0.1531 = \0
Volume Block 3 = $0 \times \$0.1631 = \0
Quarterly Bill = \$134.20

Example: A residential premise (e.g. a summer holiday beach) is read quarterly. On this occasion, the customer has the following consumption pattern.

	Meter reading period (days)	Consumption kWh
Quarter 1	90	1000
Quarter 2	88	0
Quarter 3	93	0
Quarter 4	95	0

The meter reading at the start of quarter 1 is 240,000 kWh and the reading at the end of the fourth quarter is 241,000 kWh.

Calculation:

Quarter 1

Equivalent Daily Consumption = consumption *divided by* the number of days in the read = $1000/90 = 11.11$.
Fixed Charge = $90 \times \$1.525 = \137.2500
Volume Charge Block 1 = $2.74 \times \$0.00 \times 90 = \0
Volume Block 2 = $(11.11 - 2.74) \times \$0.1531 \times 90 = \115.33
Volume Block 3 = $(0) \times \$0.1631 \times 90 = \0
Quarterly Bill = \$252.58

Quarter 2

Equivalent Daily Consumption = consumption *divided by* the number of days in the read = $0/88 = 00.00$.
Fixed Charge = $88 \times \$1.525 = \134.2000
Volume Charge Block 1 = $(0) \times \$0.00 \times 90 = \0
Volume Block 2 = $(0) \times \$0.1531 \times 90 = \0
Volume Block 3 = $(0) \times \$0.1631 \times 90 = \0
Quarterly Bill = \$134.20

Quarter 3

Equivalent Daily Consumption = consumption *divided by* the number of days in the read = $0/93 = 00.00$.
Fixed Charge = $93 \times \$1.525 = \141.8250
Volume Charge Block 1 = $(0) \times \$0.00 \times 90 = \0
Volume Block 2 = $(0) \times \$0.1531 \times 90 = \0
Volume Block 3 = $(0) \times \$0.1631 \times 90 = \0
Quarterly Bill = \$141.83

Quarter 4

Equivalent Daily Consumption = consumption *divided by* the number of days in the read = $0/95 = 00.00$.
Fixed Charge = $95 \times \$1.525 = \144.8750
Volume Charge Block 1 = $(0) \times \$0.00 \times 90 = \0
Volume Block 2 = $(0) \times \$0.1531 \times 90 = \0
Volume Block 3 = $(0) \times \$0.1631 \times 90 = \0
Quarterly Bill = \$144.88

Annual total \$673.49.

Appendix 4: Ergon Energy's kVA calculation methodology

The four metering data quadrants recorded by interval meters are referred to as:

- kWh_{Load} the real component of load power consumed by the customer over a given time period
- kWh_{Gen} the real component of power generated by the customer over a given time period
- kVA_{rhLag} (kilovolt-amperes-reactive-hours) is a measure of the reactive power which exists when the current and voltage are out of phase, where the current waveform is lagging the voltage waveform
- kVA_{rhLead} is a measure of the reactive power which exists when the current and voltage are out of phase, where the current waveform is leading the voltage waveform.

The values given above are converted to a kW or kVA demand based measure by multiplying the kWh or kVArh 30 minute interval data by a factor of two (2), to calculate the kWh consumption for one (1) hour. This resulting value equates to the kW real power or kVA reactive power demand on the network during that half hour period.

Ergon Energy will apply the following equation to calculate the kVA value used to determine the DUOS capacity and actual demand charges for ICCs:

$$kVA = \sqrt{(kW_{Load})^2 + (kVA_{rhLag})^2} \text{ for each half hour interval}$$

Where:

kVA (kilovolt-amperes) is the measure of instantaneous demand placed on the electricity network, for which Ergon Energy must provide network capacity.

The calculation is applied for each half hour interval of the month to determine the maximum value of kVA demand in that month.

Ergon Energy will also apply the above equation to calculate the kVA value used to determine the TUOS capacity charge for ICCs.

In the application of this formula:

- Where an interval meter records data at 15 minute intervals, adjacent intervals will be summed to 30 minute intervals.
- Within a 30 minute interval, there is no netting-off of interval data as between kWLoad and kWGen, or between kVArhLag and kVArhLead. The half hourly interval values of kWLoad and kVArhLag are used directly as described above to determine the kVA demand.
- Where Ergon Energy has agreed to the summation of the customer's demand at more than one metering installation for the purposes of determining a diversified demand charge, the billable demand is the vector summation of kWLoad and kVArhLag at the individual metering installations. That is, the chargeable demand is calculated for each interval as:

$$kVA = \sqrt{(kW_{Load}^{Meter\ 1} + kW_{Load}^{Meter\ 2} + \dots)^2 + (kVA_{rhLag}^{Meter\ 1} + kVA_{rhLag}^{Meter\ 2} + \dots)^2}$$

- The kVA demand is determined for each half hour interval in the billing period and the maximum value during the billing period is used for billing purposes.

Glossary

Abbreviations

ACS	Alternative Control Services
AD	Authorised demand
AEMO	Australian Energy Market Operator
AER	Australian Energy Regulator
CAC	Connection Asset Customer
DLF	Distribution Loss Factor
DNSP	Distribution Network Service Provider
DUOS	Distribution Use of System
EG	Embedded Generator
Ergon Energy	Ergon Energy Corporation Limited
GWh	Gigawatt hour
IBT	Inclining Block Tariff
ICC	Individually Calculated Customer
kV	Kilovolt
kVA	Kilovolt-ampere
kW	Kilowatt
kWh	Kilowatt hour
Law	National Electricity Law
LGA	Local government area
MWh	Megawatt hour
NEM	National Electricity Market
NER	National Electricity Rules
NMI	National Metering Identifier
p.a.	Per annum
POA	Price on Application
PV	Photovoltaic
QCA	Queensland Competition Authority
SAC	Standard Asset Customer
SCS	Standard Control Services
TNCP	Transmission Network Connection Point
TNI	Transmission Node Identifier
TNSP	Transmission Network Service Provider
TOU	Time-of-Use
TUOS	Transmission Use of System

Definitions

ACS – Street Lighting Services	A type of Alternative Control Service. Relates to the installation, operation and maintenance of street lighting assets owned by Ergon Energy. Ergon Energy recovers our costs in providing this service through a daily street lighting charge which we bill to retailers.
Actual demand charge	A type of charge (charging parameter) included in Ergon Energy's network tariff structures to signal the effect demand has on the shared network and system augmentation. The demand used in the calculation of the charge is the maximum demand recorded in any half hour period each month.
Alternative Control Service	A distribution service provided by Ergon Energy that the AER has classified as an Alternative Control Service under the NER. Includes Fee Based Services, Quoted Services and ACS – Street Lighting Services.
Australian Energy Regulator (AER)	The AER is an independent statutory authority that is part of the Australian Competition and Consumer Commission. The AER is responsible for the economic regulation of electricity networks in the NEM. It also monitors the wholesale electricity and gas markets and is responsible for compliance with and enforcement of the National Electricity Law and NER, National Gas Law and Rules, and the National Energy Retail Law and Rules.
Business customer	Means a customer who is not a residential customer (as defined in the Queensland Electricity Industry Code).
Capacity charge	A type of charge (charging parameter) included in Ergon Energy network tariff structures. The capacity charge is reflective of the costs associated with the network capacity required by a customer on a long term basis. It is similar to the actual demand charge, but more effectively takes into account the impact low load factor customers have on system augmentation. The demand used for the calculation of the charge is the AD or, if there is no AD, an annual maximum demand.
Capital contribution	A capital contribution is a prepayment for the provision of direct control services. A capital contribution may be charged to a customer if the new connection or modification for an existing connection is required to the network to accommodate the connection/modification. Ergon Energy's Capital Contribution Policy sets out circumstances in which a capital contribution may be required and details how the capital contribution to be charged to a customer is calculated.
Charging parameter	The constituent elements of a tariff (as defined in the NER).
Connection	The physical link to or through a transmission network or distribution network.
Connection Asset Customer (CAC)	Means a customer classified as a CAC in accordance with definitions in our Pricing Proposal. Typically reflect those customers with required capacity above 1,500 kVA, or with electricity consumption greater than 4 GWh (but less than 40 GWh) per year.
Connection assets	Those components of a transmission or distribution system which are used to provide connection services. Connection assets are those assets required to connect an electrical installation to the shared network and are all the assets from the connection point back up to and including the network coupling point.

Connection point	The agreed point of supply established between Network Service Provider(s) and another Registered Participant, Non-Registered Customer or franchise customer.
Customer	A person or entity that receives, or wants to receive a supply of electricity for a premises, or any other distribution service from Ergon Energy.
Dedicated connection assets	Assets that are dedicated to a network user of any size. In most circumstances these are 'connection assets'. This is because no other party is using the assets and it enables the user to pay for exactly the capacity and style of assets they seek to have. In some cases, dedicated connection assets may later be shared with another customer in which case the network coupling point is moved further downstream.
Demand	The amount of electricity energy being consumed at a given time measured in either watts (W) or volt amperes (VA). The difference between the two is the power factor.
Direct control service	Distribution services subject to economic regulation by the AER under the NER. Direct control services are further subdivided into Standard Control Services and Alternative Control Services.
Distribution Loss Factor (DLF)	The factor applied to customers' metered energy consumption to determine total energy purchases at the transmission connection point that account for losses in the distribution network.
Distribution network	The electrical system used to transport electricity from the high voltage transmission network connection point to distribution network users.
Distribution Use of System (DUOS) charge	Component of the network tariffs which covers costs associated with connection services and/or use of the distribution network for the conveyance of electricity (i.e. Standard Control Services).
East Zone	Those areas where the network users are supplied from the distribution system connection to the national grid and have a relatively low distribution cost to supply. The local government areas covered by the East Zone are located in this Guide.
Embedded Generator (EG)	Means a network user classified as an EG in accordance with the definition in our Pricing Proposal. EGs are those network users that export energy into the distribution system, except for those customers with micro generation facilities that have been classified as a SAC (such as small scale PV generators).
Energy	The amount of electricity consumed by a consumer (or all customers) over a period of time. Energy is measured in terms of watt hours (Wh), kilowatt hours (kWh), megawatt hours (MWh) or gigawatt hours (GWh).
Fee Based Service	A type of Alternative Control Service which Ergon Energy undertakes at the request of an identifiable customer, retailer or appropriate third party which is in addition to our main network, connection and metering services and is levied as a separate charge. Fee Based Services have a fixed fee as the costs of providing the service (and therefore price) can be assessed in advance of the service being requested by a customer or retailer.
Final Distribution Determination	The AER's Final Distribution Determination sets the revenue and pricing control regime that Ergon Energy must comply with for the current regulatory control period.

Fixed charge	A type of charge (charging parameter) included in Ergon Energy network tariffs which reflects the costs associated with connection assets (entry and exit services) and network user management services. For those customers with consumption less than 100 MWh per annum (SACs <100 MWh p.a.), the fixed charge also recovers a portion of the shared network costs. The charge is levied on a fixed dollar amount per day.
Gigawatt hour (GWh)	1,000,000 kilowatt hours.
High voltage (HV)	Refers to parts of the network that are 11 kV or above.
Inclining Block Tariff (IBT)	A type of network tariff where the price per kWh increases as consumption thresholds are crossed during a particular time period.
Individually Calculated Customer (ICC)	Means a customer classified as an ICC in accordance with definitions in our Pricing Proposal. Typically reflects those customers with electricity consumption greater than 40 GWh per year, or where a customer's circumstances and connection arrangement mean that average prices are meaningless or inappropriate (e.g. only two or three customers in a supply system, or customer connected close to a TNCP).
Isolated generation	Those areas supplied from Ergon Energy's isolated generation assets, except for the Mount Isa system. Includes communities in Western Queensland, the Gulf of Carpentaria, Cape York, various Torres Strait Islands, Palm Island and Mornington Islands. These areas are not subject to economic regulation by the AER, but are regulated by the Queensland Government.
kVA	1,000 Volt-Ampere which is a measure of the apparent power flow which is a measure of the total capacity required to supply a customer's load.
kW	1,000 Watts which is a measure of the real component of power being consumed by the consumer's load.
Large customer	Are Individually Calculated Customers (ICCs), Connection Asset Customers (CACs) or Embedded Generators (EGs).
Large Customer Connection arrangements	Refers to the arrangements applying from 1 July 2010, where new or augmented connection assets are paid for or contributed by the large customer (i.e. not included in the network tariff).
Large Customer Connection service	Is a type of Quoted Service which relates to the design and construction of connection assets for large customers.
Load factor	Measure of the percentage of time a load is used in any given period. Loads used 24 hours per day, 7 days a week have a load factor of 1 or 100%.
Low voltage (LV)	Refers to the sub 11 kV network.
Maximum demand	The maximum demand recorded at a customer's individual meter or the maximum demand placed on the electrical distribution network system at any time or at a specific time or within a specific time period, such as a month. Maximum demand is an indication of the capacity required for a customer's connection or the electrical distribution network.
Megawatt hour (MWh)	1,000 kilowatt hours

Mount Isa Zone	Those areas supplied from the isolated Mount Isa system. This zone is not connected to the national grid and would normally be excluded from the application of the NER. However, under the <i>Electricity – National Scheme (Queensland) Act 1997</i> , the Queensland Government has transferred responsibility for the economic regulation of the Mount Isa-Cloncurry supply network to the AER. The local government areas covered by the Mount Isa Zone are located in this Guide.
National Electricity Market (NEM)	The interconnected electricity grid covering Queensland, New South Wales, Victoria, Tasmania, South Australia and the Australian Capital Territory.
National Electricity Rules (NER)	Rules made under the National Electricity Law which govern the operation of the NEM.
National Metering Identifier (NMI)	A unique number assigned to each metering installation.
Network capacity	The maximum demand (kW) that the distribution network can provide for at any one time.
Network coupling point	The point at which connection assets join a distribution network, used to identify the distribution service price payable by a connection customer.
Network tariff	Refers to the price (or tariff) that Ergon Energy sets to recover costs associated with the customer's connection and use of the distribution and transmission network. Network tariffs comprise DUOS and TUOS components.
Network user	There are four network user groups included in Ergon Energy's network tariff structures – Individually Calculated Customers (ICCs), Connection Asset Customers (CACs), Standard Asset Customers (SACs) and Embedded Generators (EGs). For the purposes of our network pricing documents, the term 'network user' refers to both a 'customer' and an 'EG'.
Premises	Means premises owned or occupied by the customer.
Primary tariff	The main network tariff applicable to a customer's premises. In some circumstances an additional (or secondary) tariff may apply to particular loads that are separately metered at a customer's premises (e.g. our Volume Controlled and Volume Night Controlled network tariffs which are available for customers who have appliances eligible to be controlled by network equipment).
Quoted Service	A type of Alternative Control Service. Similar to Fee Based Services, but they are priced on application as the nature and scope of these services is variable and the cost (and therefore price) are specific to the individual retailer's or customer's needs.
Regulatory control period	The regulatory control period is a five (5) year period set down by the AER. The current regulatory control period is 2010–11 to 2014–15.
Regulatory year	Is a specific financial year within a regulatory control period.
Residential customer	Means a customer who acquires electricity for domestic use (as defined in the Queensland Electricity Industry Code).
Solar Bonus Scheme	A Queensland Government program that requires Ergon Energy to pay eligible customers for the surplus electricity generated from roof-top solar PV systems that is exported to the Queensland grid.

Standard Asset Customer (SAC)	Means a customer classified as a SAC in accordance with the definition in our Pricing Proposal. Typically reflects those customers with annual electricity consumption below 4 GWh per year. Includes customers with micro generation facilities (such as small scale PV generators) that have a similar service connection and usage profile as other SACs without such facilities.
Standard Control Service	A distribution service provided by Ergon Energy that the AER has classified as a Standard Control Service under the NER. Includes core network, connection and metering services associated with the access and supply of electricity to customers. Ergon Energy recovers our costs in providing Standard Control Services through the DUOS component of network tariffs which are billed to retailers.
Street lights – Major	Standard major street lights are 100, 150, 250 and 400 watt High Pressure Sodium vapour lights and include any other non-standard or obsolete street lights that would be replaced with any of the above lights.
Street lights – Minor	Standard minor street lights are 50, 80 and 125 watt Mercury Vapour, 70 watt High Pressure Sodium vapour lights, and compact fluorescent lights of similar luminescence and include any other non-standard or obsolete street lights that would be replaced with any of the above lights.
Summer	The months of December, January and February.
Tariff class	A class of customers for one or more direct control services who are subject to a particular tariff or particular tariffs (as predefined in the NER).
Threshold demand	The amount by which a SAC >100 MWh p.a. customer's metered monthly actual kW maximum demand is adjusted for the purposes of calculating the demand component of their network tariff. The actual demand charge tariff charging parameter (\$/kW/month) is applied to the higher of metered monthly demand less the applicable threshold, or zero. The threshold adjustment is effective for 2014–15 SAC >100 MWh p.a. network tariffs. The adjustment was introduced to avoid over-recovery of revenue arising from increasing the fixed charge to offset the removal of the minimum monthly demand charge in 2014–15.
Time-of-Use (TOU)	A type of network tariff where the price per kWh varies according to when the consumption occurs. The TOU tariff applies a different price during Peak, Shoulder and Off-Peak periods.
Transmission Use of System (TUOS) charge	Component of the network tariff which passes through costs associated with use of the transmission network. This includes designated pricing proposal charges as defined under the NER.
Unmetered	A customer who takes supply where no meter is installed at the connection point.
Volume charge	A type of charge (charging parameter) included in Ergon Energy network tariffs which in part recovers costs that have been allocated on a postage stamped basis. The volume charge is calculated using the customer's metered energy (kWh) consumption and may be based on a flat rate, an inclining block or TOU charging structure (depending on the customer's applicable network tariff).
West Zone	Those areas outside the East Zone and connected to the national grid, which have a significantly higher distribution cost of supply than the East Zone. The local government areas covered by the West Zone are located in this Guide.

Customer Service

13 10 46

7.00am – 6.30pm, Monday to Friday

Faults Only

13 22 96

24 hours a day, 7 days a week

Life-Threatening Emergencies Only

Triple zero (000) or 13 16 70

24 hours a day, 7 days a week

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Ergon Energy Queensland Pty Ltd ABN 11 121 177 802

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